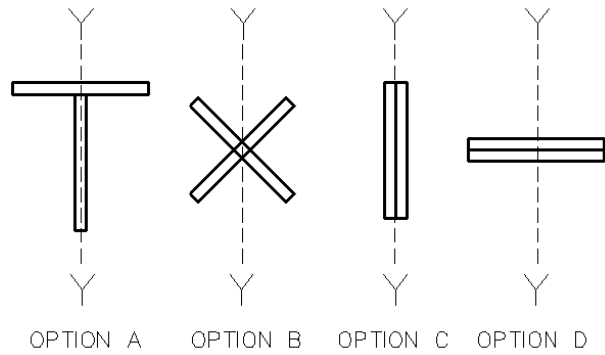


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001

86. Two 1 inch by 12 in plates are used to form a cross section. The four shapes formed are shown as follows



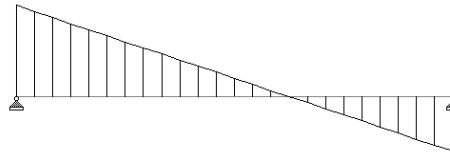
Which shape will have the highest flexural stiffness about Y-Y axis?

- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

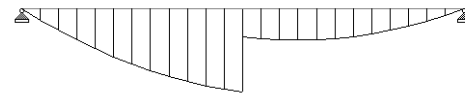
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89. For a simply supported beam, the shear force diagram and the bending moment diagram are shown below. Which of the following best describes the loading condition in the beam?
Moment diagram is drawn on the tension side of the beam.

Shear Force



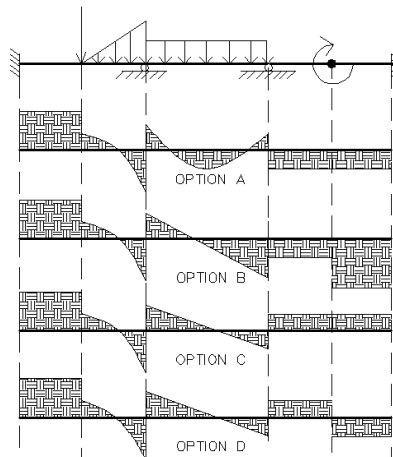
Bending Moment



- (A) Uniformly distributed load with a concentrated load at the midspan
- (B) Uniformly distributed load with anti-clockwise moment at the midspan
- (C) Uniformly distributed load with clockwise moment at the midspan
- (D) Uniformly varying load with a couple at the midspan

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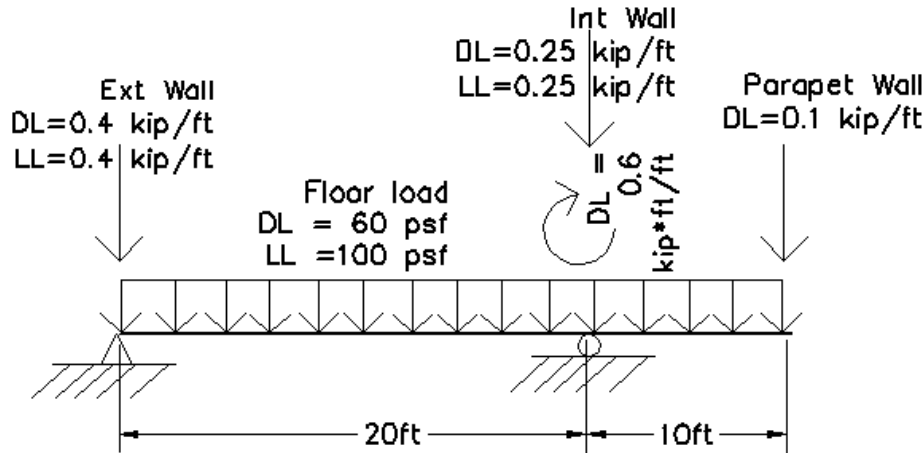
91. For the given three span beam with a hinge at the center of the third span, what is the correct shear force diagram?



- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

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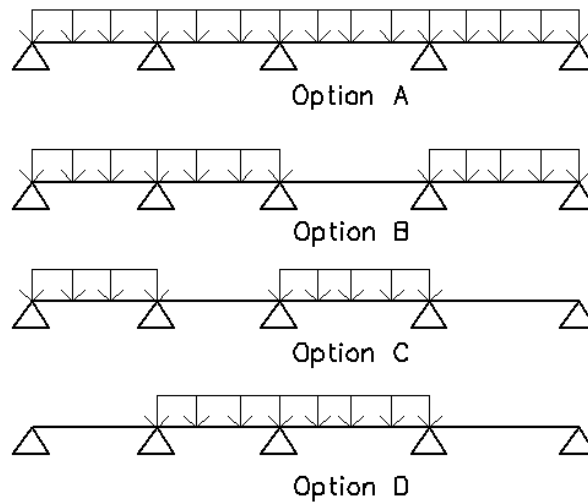
- 138.** Loads acting on a simply supported floor beam is shown below. The floor beams are spaced 8 ft c/c. What is the maximum service load shear acting on the beam ?



- (A) 2.08 kips
(B) 8.96 kips
(C) 15 kips
(D) 16.64 kips

P0
005

- 143.** Four span continuous beam is subjected to uniformly distributed loadings in different spans as shown below. Which loading configuration produces maximum negative moment in an internal support?



- (A) Option A
(B) Option B
(C) Option C
(D) Option D

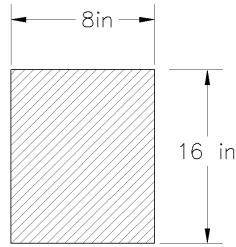
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- 189.** Two beams with identical cross sections and lengths are manufactured with two materials. One of them is manufactured with gray cast iron and another is manufactured with mild steel. Both of them are fixed in two ends identically and identical uniform loading is applied. Which of the following is true about their deflections?

- (A) Gray cast iron beam will deflect less than mild steel beam,
(B) Mild steel beam will deflect less than gray cast iron beam,
(C) For identical cross section, loading and statically determinate conditions, the deflections will be equal for both materials.
(D) Deflections will depend upon the allowable yield stresses of both materials.

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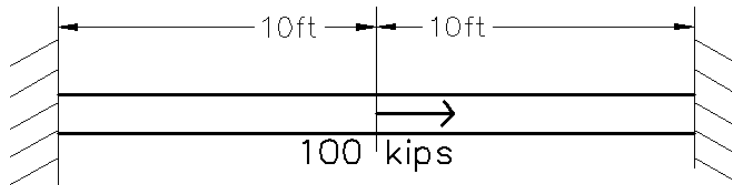
- 160.** A rectangular beam is subjected to a compressive load of 150 kips. In addition to the compressive force it is subjected to biaxial bending. The dimensions of the beam is 8 in X 16 in. What is the maximum normal stress acting on the beam section?
 Moment about stronger axis = 100 kip*ft
 Moment about weaker axis = 50 kip*ft



- (A) 1.2 ksi
 (B) 5.6 ksi
 (C) 8.2 ksi
 (D) 12.1 ksi

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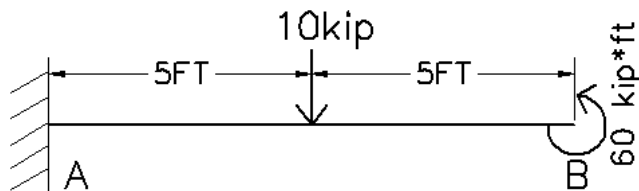
- 171.** A fixed-fixed beam with area 11 in^2 is subjected to an axial load at the center of the span. The beam was installed at 32°F with no room for expansion and the ambient temperature at service is 102°F . What is the total combined compressive stress acting in the beam section? Modulus of Elasticity = 29000 ksi Coefficient of thermal expansion = $0.000007 / ^\circ\text{F}$.



- (A) 4.54 ksi
 (B) 9.09 ksi
 (C) 18.76 ksi
 (D) 20.34 ksi

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009

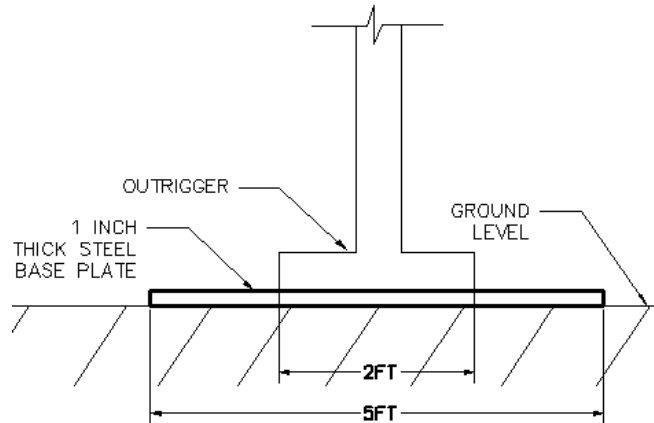
- 191.** What is the deflection at the free end of the cantilever beam due to the loads shown below? $I = 50 \text{ in}^4$ $E = 29000 \text{ ksi}$



- (A) 1.2 in
 (B) 2.33 in
 (C) 3.1 in
 (D) 4.8 in

P0
010

- 185.** An outrigger from a mobile crane is laid on a 5ftX5ft X1in steel base plate. The foot of the outrigger is 2ftX2ft. The maximum concentric axial thrust from the outrigger is 50 kips. What is the maximum deflection of the tip of the steel base plate? $E = 29000$ ksi



- (A) 0.015 in
(B) 0.03 in
(C) 0.05 in
(D) 0.075 in

P0
011

- 8.** A timber beam at the roof of a building supports a parapet wall along its length. The wall is 12 ft long and 4 ft tall. The wall is constructed with hard brick masonry and is 9 inches thick. What is the factored dead load to be used in combination with live loads, due to the masonry's weight acting on the beam?

- (A) 390 lb/ft
(B) 468 lb/ft
(C) 561 lb/ft
(D) 673 lb/ft

P0
012

- 12.** A two-story gymnasium is to be built in New York. The ground floor slab is on grade and the first floor will be supported by steel beams. The tributary area for an interior beam of length 40 ft is 1250ft². What is the minimum live load acting on the interior beam? Apply reduction factors as applicable.

- (A) 6.25 kips/ft
(B) 3.125 kips/ft
(C) 2.5 kips/ft
(D) 1.5 kips/ft

P0
013

- 20.** ASCE 7-16 Table 4.3-1 provides minimum distributed live loads and minimum concentrated live loads to be used for design for structures. Which of the following is not true about concentrated loads?

- (A) Concentrated live loads are assumed to act uniformly over an area of 0.58 m²
(B) ASCE 7-16 specified concentrated live loads are summed with minimum uniformly distributed live loads to create the most severe effect in the structural member
(C) Live load reduction factors cannot be applied to concentrated loads.
(D) Concentrated loads are usually critical over uniformly distributed loads during slab design.

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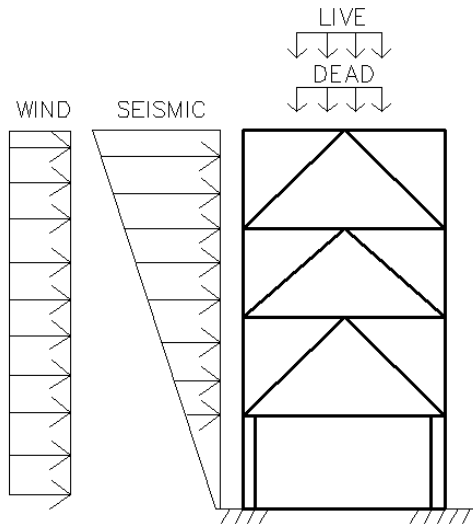
- 81.** The maximum bending moments on a roof beam due to different load cases per ASCE 7-16 are as follows:
- Loads from slab and parapet wall = 40 kip*ft
 - Load from a water tank = 75 kip*ft
 - Roof live load = 5 kip*ft
 - Rain load = 10 kip*ft
 - Snow load = 8 kip*ft
 - Vertical seismic load = 5 kip*ft
 - Horizontal seismic load = 25 kip*ft

What is the governing moment for allowable stress design?

- 112 kip*ft
- 125 kip*ft
- 136 kip*ft
- 137 kip*ft

P0
015

- 82.** For the building shown below, which of the following load combinations must be considered to calculate the governing factor of safety for stability against overturning?



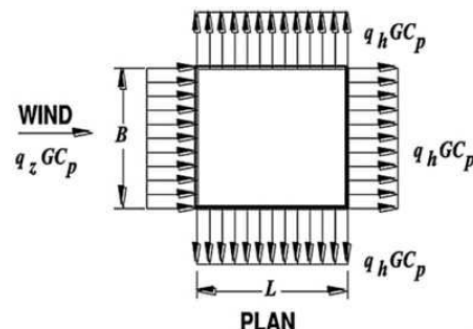
- Dead + Live + Wind
- Dead + Seismic + Wind
- Dead + Seismic or Wind
- Dead + Live + Seismic + Wind

P0
016

- 33.** Using the directional procedure, compute the design wind pressure in the windward walls of a rigid enclosed 1 story building located in Exposure C terrain. The basic wind speed of the area is 100 mph. Neglect any ground elevation and topographic effects.

$L = 30\text{ft}$
 $B = 30\text{ft}$
 Height of wall $z = h = 10\text{ft}$

- 9.25 psf, 15.91 psf
- 15.91 psf, -9.25 psf
- 9.25psf, -15.91 psf
- 9.25psf, -15.91 psf



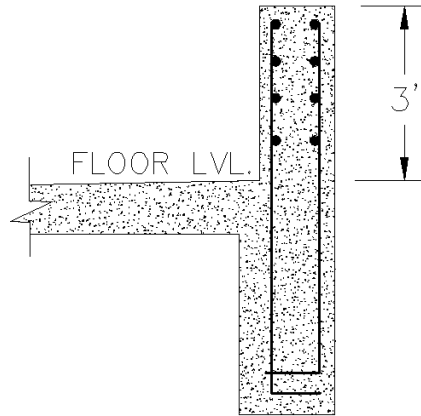
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017

- 34.** A curtain wall is to be replaced in an Exposure C zone. The basic wind speed of the area specified by ASCE 7-16 chapter 26 is 100 mph. The specifications of the material of the curtain wall only list allowable stress values and the strength of the wall against wind loads cannot be checked in accordance with LRFD methodology. What is the basic wind speed to be used in ASD design of the wall?

(A) 100 mph
(B) 77.5 mph
(C) 85 mph
(D) 90 mph

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018

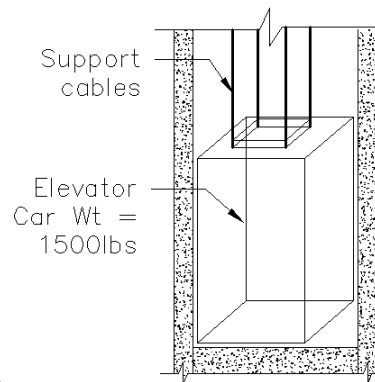
- 61.** Given detail below is the section of a barrier in a 2nd floor parking lot. The barrier system is designed per the provisions of ASCE 7-16 for impact by passenger vehicles. What is the maximum factored design moment acting at the slab level?



(A) 9 kip*ft
(B) 13.5 kip*ft
(C) 14.4 kip*ft
(D) 21.6 kip*ft

P0
019

- 63.** An elevator car weighs 1500 lbs and is rated for a total passenger weight of 2000 lbs. The elevator is supported by hoisting cables on four corners of the car as shown in the sketch below. What is the minimum safe working load limit of each cable supporting the car? The impact factor to be used for design is 100% .



(A) 1 kip
(B) 2 kips
(C) 3 kips
(D) 4 kips

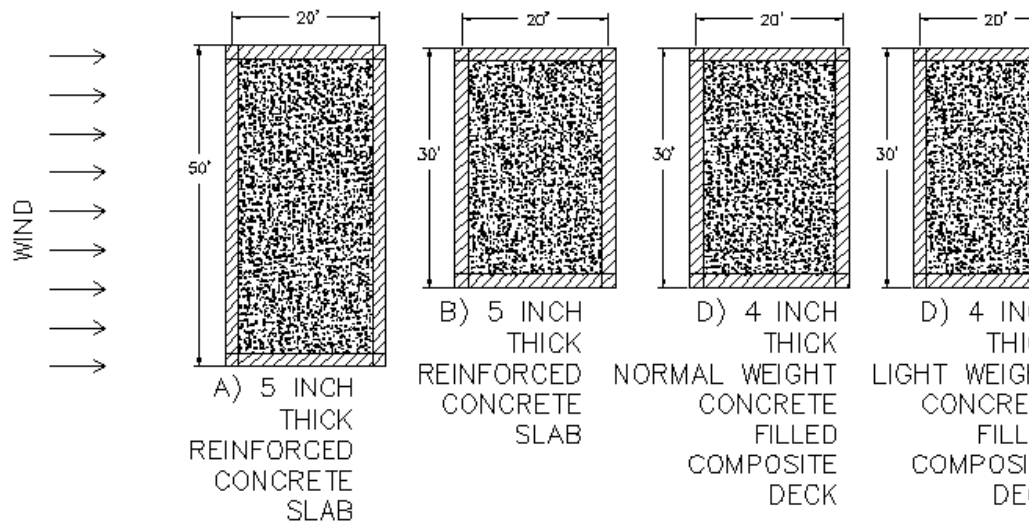
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020

64. A structural designer wants to account for impact loads in a semi-commercial building. Which one of the following provisions is not true per ASCE 7-16?

- (A) Loads due to unusual vibrations should be accounted for while designing machine foundations. These load are not provided in ASCE 7-16 and should be provided by the equipment manufacturer,
- (B) The operating weights of motor driven light machinery should be increased by 20%,
- (C) The vertical impact due to wheel loads of pendant operated bridge crane is taken as 10% of the wheel load,
- (D) The equipment/moving load factored with appropriate impact factors need not have the additional 1.6 live load factor for LRFD design.

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021

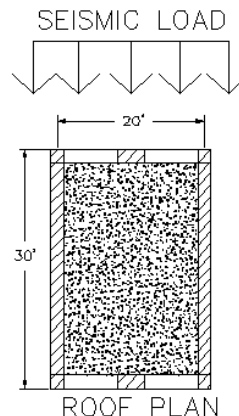
26. The plans below show Roof slabs for four different buildings. Which of the following cannot be idealized as a rigid diaphragm for analysis under wind loads per ASCE 7 -16?



- (A) A
- (B) B
- (C) C
- (D) D

P0
022

36. The plan below shows the roof plan of a one story building. The roof is a concrete slab, supported by masonry walls on longer sides and columns on shorter sides as shown by the hatches. During a seismic event, the lateral load acting at the roof slab level is 200 lb/ft and is along the direction as shown below. What is most nearly the loads shared by each wall?

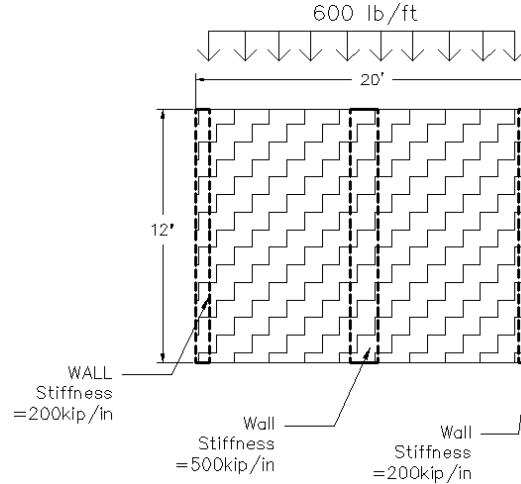


- (A) 1 kips
- (B) 2 kips
- (C) 3 kips
- (D) 4 kips

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023

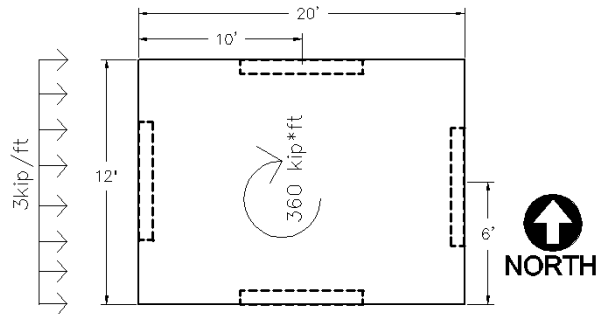
75. The plan shown below is a light gauge steel deck floor and is supported by three walls underneath. The stiffness of the walls supporting the floor is indicated in the figure. A lateral load of 600 lb/ft is applied on the diaphragm. What is the shear load per unit length acting on the central wall?

- (A) 500 lb/ft
(B) 600 lb/ft
(C) 666.66 lb/ft
(D) 700 lb/ft

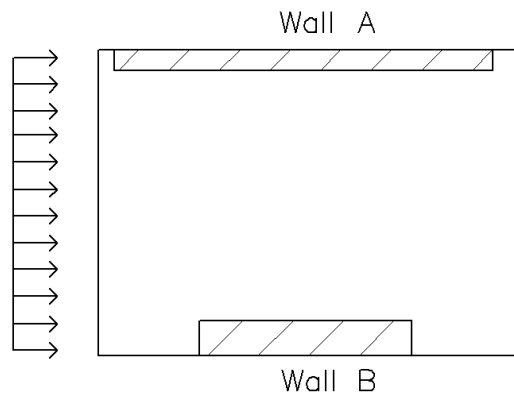
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024

79. The rigid diaphragm shown below has a lateral load and a torsional moment acting at the center. The center of rigidity is at the geometric center of the diaphragm. The in-plane stiffness of each wall is equal. What is the maximum shear load on an individual wall?

- (A) 18 kips
(B) 24 kips
(C) 26 kips
(D) 30 kips

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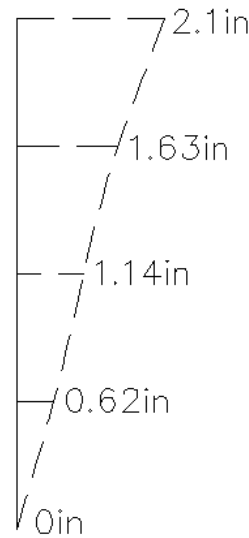
345. The plan of a concrete floor diaphragm is shown below. The diaphragm is supported by two walls and subjected to seismic loads. The story drift of Wall A is analyzed as 0.45 inch and the story drift of wall B is analyzed as 0.6 inch. The in plane deflection of the diaphragm itself is 1.1 inches. Which of the following best describes the diaphragm?



- (A) Rigid
(B) Rigid and torsionally irregular
(C) Flexible and torsionally irregular
(D) Flexible

P0
026

- 182.** A four story steel special moment resisting frame building is identified as an essential infrastructure and a seismic design category C. The elastic deflections of the floors are as follows. What is the maximum design story drift per ASCE 7-16?



- (A) 2.1 in
(B) 2.27 in
(C) 2.5 in
(D) 2.78 in

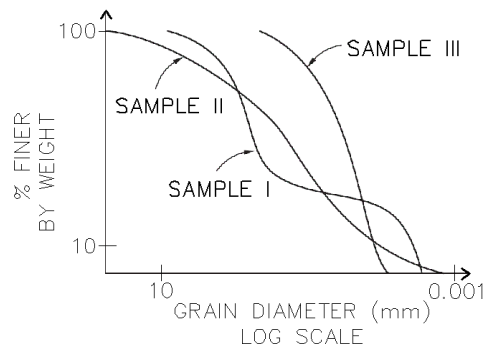
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027

- 183.** The average story height of a three story steel frame building is 11ft. It is identified as Seismic Risk category III and seismic design category D. The structure has extreme torsional irregularity. What is the allowable story drift per ASCE 7-16?

- (A) 2 in
(B) 2.64 in
(C) 3 in
(D) 3.2 in

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028

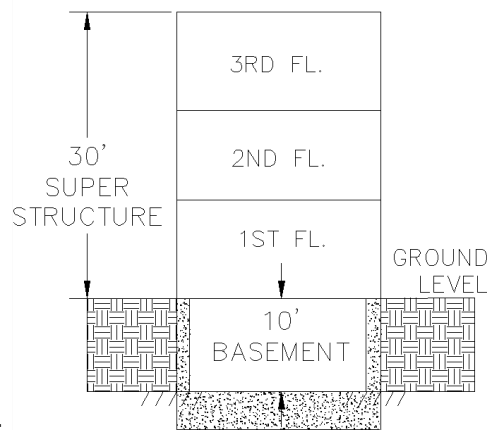
- 245.** The graph below shows particle size distribution for three soil samples I, II and III. Which of the following denotes the correct grading of the samples?



- | | | |
|---------------------|-----------------|------------------|
| (A) I - Well graded | II- Gap graded | III- Uniform |
| (B) I - Gap graded | II- Well graded | III- Uniform |
| (C) I - Uniform | II- Well graded | III- Gap graded |
| (D) I - Gap graded | II- Uniform | III- Well Graded |

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029

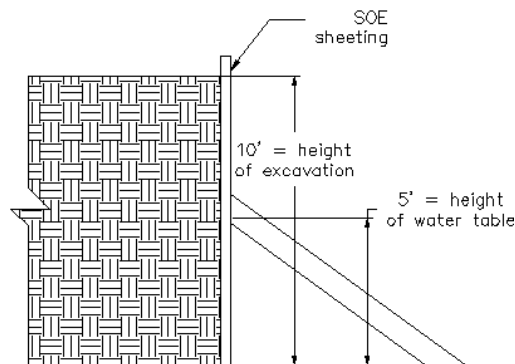
65. A three story building shown below has a 10 ft deep basement parking level. 12 inch thick reinforced concrete foundation walls extend from the mat to the 1st floor slab level. The soil retained by the foundation wall is well graded gravel-sand mixes. Calculate the total lateral load exerted by the soil on the East foundation wall. The total length of the East wall is 40 feet. Water table can be assumed below the foundation mat.



- (A) 60 kips
(B) 90 kips
(C) 120 kips
(D) 150 kips

P0
030

67. What is the total horizontal force acting per unit length of the SOE sheeting as shown in the figure below? The height of excavation is 10ft and the water table is at 5ft. The active earth pressure coefficient is 0.33. The dry density of the soil is 110 pcf and the submerged density is 65 pcf.



- (A) 1.1 kips/ft
(B) 1.3 kips/ft
(C) 1.9 kips/ft
(D) 2.4 kips/ft

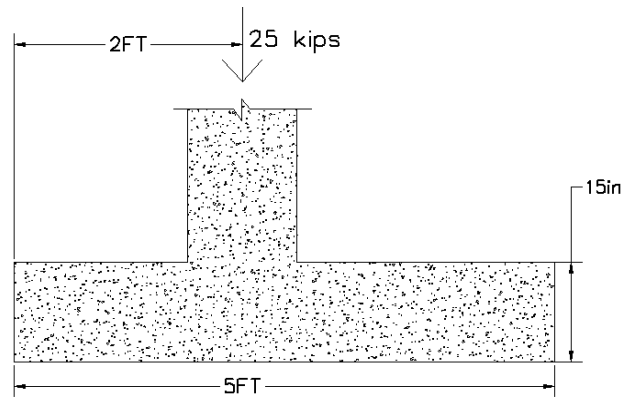
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031

73. Which of the following is not correct about lateral earth pressure?

- (A) Lateral earth pressure coefficient is the ratio of horizontal stress to vertical stress exerted by the soil,
(B) The strain required to achieve passive pressure is significantly larger than the strain required to reach active pressure,
(C) Active earth pressure cannot be negative regardless of properties of soil,
(D) The lateral pressure exerted by soil retained by bridge abutments connected to the superstructure deck above is calculated by using the coefficient of earth pressure at rest.

P0
032

- 198.** The figure below shows a 5 ft X 5ft square footing. The column load including self weight of the column acting on the footing is 25 kips. What is the maximum bearing pressure? The unit weight of concrete = 0.15 kcf.



- (A) 1 ksf
(B) 1.6 ksf
(C) 1.8 ksf
(D) 2.8 ksf

P0
033

- 224.** A concrete deadman block is used to anchor a cable that provides lateral stability to a falsework system. The maximum force acting on the cables is 5 kips and the horizontal inclination of the cable is 30 degrees. Due to site constraints, the maximum height of the anchor block is limited to 6ft. What is the minimum area of the block that can safely stabilize the system? The cable is attached at the center of the top surface of the block.

Factor of safety = 1.5,

Coefficient of friction between concrete and soil = 0.3,

Density of concrete = 150 lbs/ft³

- (A) 5 ft X 5 ft
(B) 6 ft X 6 ft
(C) 7 ft X 7 ft
(D) 8 ft X 8 ft

P0
034

- 361.** A 16 in X 18 in steel column base plate supporting a HSS 9X8X0.5 member is centered on a concrete pedestal. The dimensions of the pedestal are 2 ft X 2 ft. The strength of the concrete is 4 ksi and the steel plate is Gr 50. What is the maximum area of the concrete bearing frustum on which the bearing load is dispersed?

- (A) 288 in²
(B) 528 in²
(C) 576 in²
(D) 1152 in²

P0
035

- 363.** A W section column is centered on a 15 in X 15 in base plate. The plate is centered on a 18inX18in concrete pedestal. The maximum concentric axial compression acting on the plate is 1000 kips. What is the minimum strength of concrete pedestal required to bear the load safely using AISC specifications?

- (A) 4.5 ksi
(B) 7.5 ksi
(C) 8.5 ksi
(D) 10.5 ksi

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036

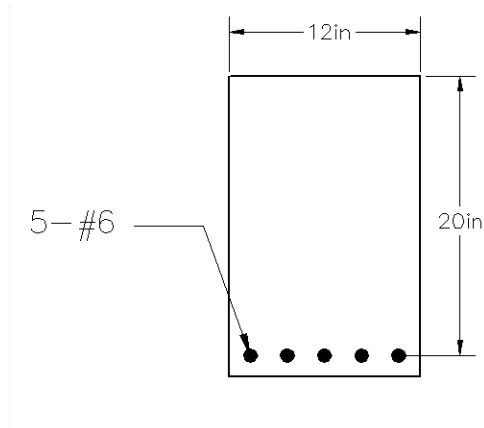
267. A 12in X 18 in simply supported non prestressed singly reinforced concrete beam section is to be designed with an effective cover of 2 inches. The strength of concrete is 3500 psi and the rebar yield strength is 60 ksi. What is the minimum number of #4 main flexural reinforcement bars that should be used following ACI recommendations?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

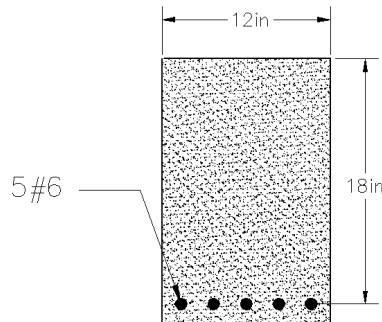
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037

268. For the section shown below, what is the strength reduction factor for flexure per ACI 318-14? Grade of steel is 60 ksi and concrete strength is 4000 psi.

- (A) 0.65
- (B) 0.75
- (C) 0.85
- (D) 0.9

P0
038

350. The factored shear to be resisted by the reinforced concrete section shown below is 51 kips. If two legged #3 stirrups are used, what is the maximum spacing of the stirrups required to satisfy the design requirements? Normal weight concrete with compressive strength $f'_c = 4$ ksi, all rebars are Grade 60.



- (A) 6 in
- (B) 9 in
- (C) 12 in
- (D) 10 in

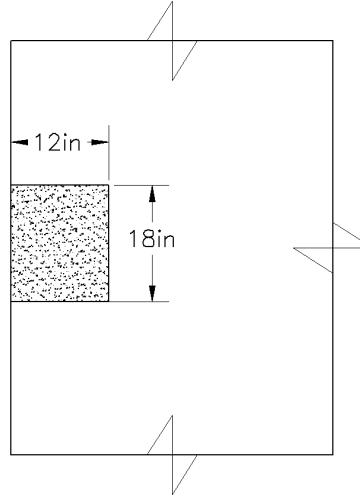
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039

354. For a one way slab with thickness 5 inches and effective depth 4 inches, what is the maximum permitted spacing of flexural reinforcement (Gr 60 deformed bars)?

- (A) 9 in
- (B) 12 in
- (C) 15 in
- (D) 18 in

P0
040

- 352.** The plan below shows a column at the edge of a floor slab. The thickness of the slab is 8in and the effective cover is 1.5in. What is the design punching shear capacity of the slab at the location around the edge column? The compressive strength of the concrete slab is 6 ksi.



- (A) 59 kips
(B) 72 kips
(C) 83 kips
(D) 103 kips

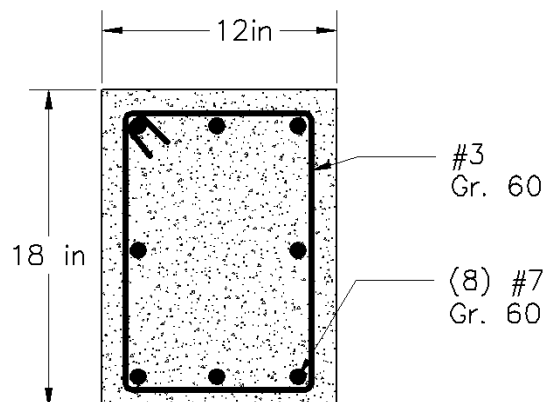
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041

- 355.** The maximum factored moment acting on a 4 inch thick one way slab is 0.7 kip*ft/ft. The compressive strength of concrete is 4 ksi and rebars used are grade 60. What is the area of reinforcement required to satisfy the design requirements per ACI 318-14? Assume effective cover for tension bars is 1in.

- (A) 0.064 in²/ft
(B) 0.086 in²/ft
(C) 0.098 in²/ft
(D) 0.12 in²/ft

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042

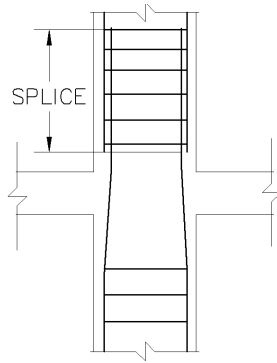
- 383.** For the column section shown below, #3 ties are used as shear reinforcements. What is the maximum permitted tie spacing per ACI 318-14?



- (A) 6 in
(B) 12 in
(C) 14 in
(D) 18 in

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043

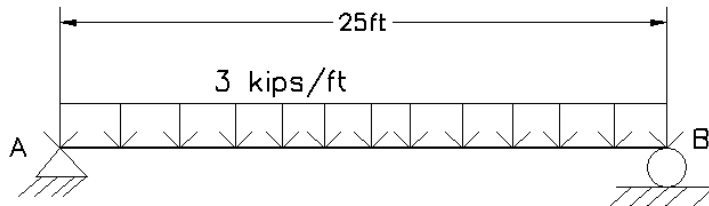
- 388.** What is the required splice length of #8 longitudinal column rebars shown below? The rebars are grade 60 and concrete $f'_c = 4$ ksi. #3 ties are spaced at 5 inches c/c in the splice location. The dimensions of the column is 24 in X 24 in and the factored load in the column is compressive.



- (A) 12 in
(B) 25 in
(C) 30 in
(D) 48 in

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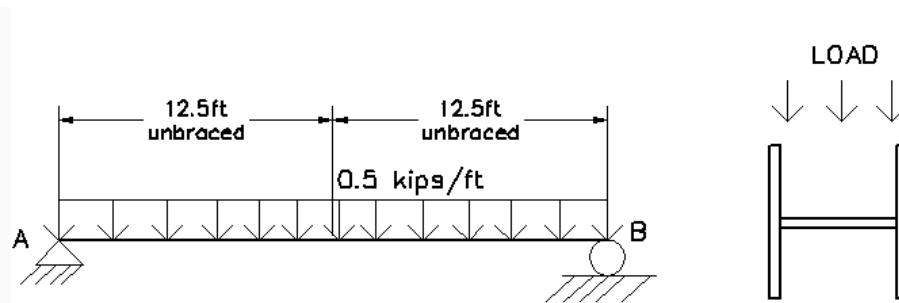
- 317.** A simply supported steel beam has a span of 25 ft and is braced at the midpoint. The beam is subjected to a uniformly distributed live load of 3 kip/ft and bends about the major axis. What is the lightest Gr. 50 section that will satisfy the flexural strength requirements?



- (A) W18X40
(B) W18X50
(C) W18X55
(D) W18X60

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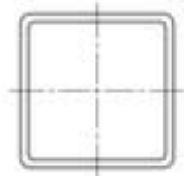
- 318.** A simply supported steel beam has a span of 25 ft and is braced at the midpoint. The beam is subjected to a uniformly distributed service live load of 0.5 kip/ft and bends about the minor axis. What is the lightest Gr. 50 section that will satisfy the flexural strength requirements?



- (A) W18X46
(B) W18X50
(C) W18X55
(D) W18X60

P0
046

- 319.** A simply supported HSS steel beam has a span of 30 ft and is used as a floor member. The beam is subjected to a service live load of 5 kip/ft and a service dead load of 1 kip/ft and bends about the major axis. What is the lightest section that will satisfy the IBC 2018 serviceability requirements?
 $E = 29000 \text{ ksi}$



- (A) HSS 20 X 20 X 1/2
- (B) HSS 20 X 20 X 5/8
- (C) HSS 20 X 20 X 3/4
- (D) HSS 22 X 22 X 3/4

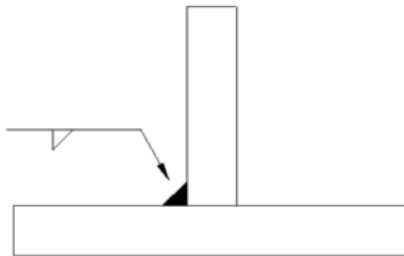
P0
047

- 359.** An unbraced W8X40 column is subjected to an axial load of 100 kips and the effective length in both directions is 12 ft. The column is also subjected to a strong axis flexure of 25 kip*ft and a weak axis flexure of 10 kip*ft. The compression flange is continuously braced against lateral torsional buckling. What is the total utilization of the column section?
 $F_y = 50 \text{ ksi}$

- (A) 60.3%
- (B) 73.9%
- (C) 82.8%
- (D) 90.1%

P0
048

- 281.** The strength of filler metal for the weld below is 70 ksi. The weld size is 3/8 inch. What is the allowable strength of the weld per unit length when it is loaded transversely to its longitudinal axis?



- (A) 5.57 kip/in
- (B) 8.35 kip/in
- (C) 11.14 kip/in
- (D) 16.7 kip/in

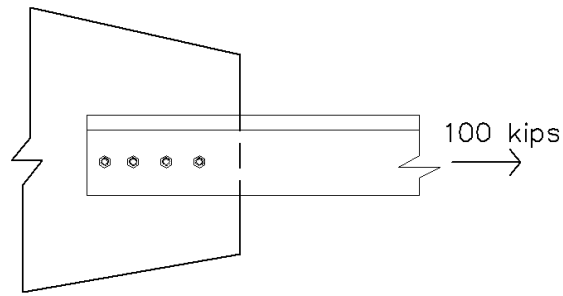
P0
049

- 435.** Which of the following statements best describes the function of designing a bolted connection to be slip critical instead of bearing connection?

- (A) Slip critical connections with high pretension and friction allow for increased shear resistance, compared to bearing type connections with same bolts,
- (B) Slip critical connections are used to prevent slip under service loads for stiffness and fatigue requirements.
- (C) Slip critical connections allows for larger bolt hole tolerances during fabrication,
- (D) Slip critical connections are designed to eliminate the need for bolt bearing and tearout checks.

P0
050

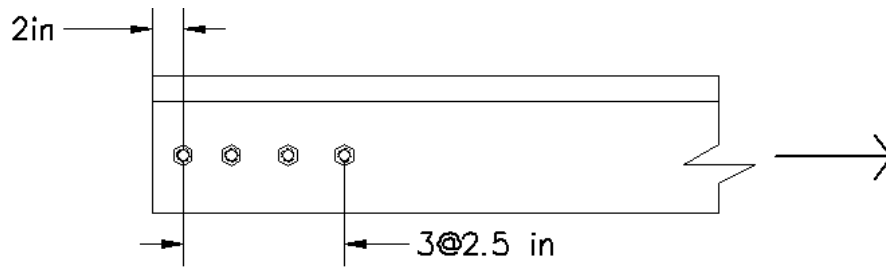
- 423.** An angle section is connected to a plate with four bolts as shown below. Assuming that the failure of the connection is gross yielding of the section, what is the most economical angle section that satisfies the design requirement? Yield strength of angle is 36 ksi. Loads shown are service loads.



- (A) L 5 X 3 X ½
- (B) L 5 X 3½ X ½
- (C) L 5 X 5 X ½
- (D) L 5 X 5 X ¾

P0
051

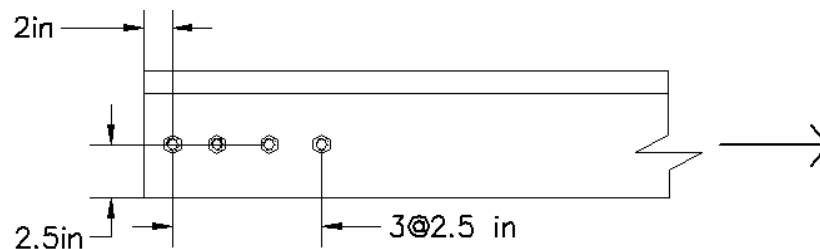
- 424.** An angle section 5X3X1/2 used as a tension member, is connected to a plate with four 0.75 inch diameter bolts as shown below. What is the governing allowable bolt bearing and tearout strength of an individual bolt if the holes are standard sizes? Yield strength of angle is 36 ksi and tensile strength is 58 ksi. Deformation at bolt hole at service load is a design consideration.



- (A) 20 kips
- (B) 22 kips
- (C) 26 kips
- (D) 28 kips

P0
052

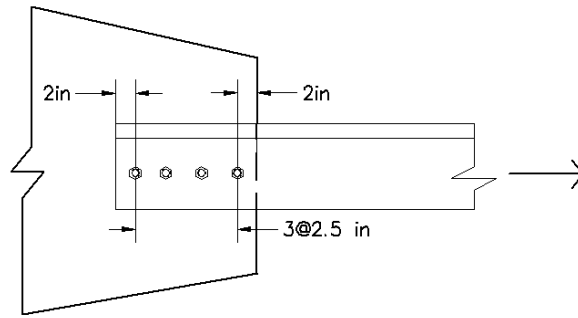
- 425.** An angle section 5X3X1/2 is used as a tension member, connected to a plate with four 0.75 inch diameter bolts as shown below. What is the allowable block shear strength of the angle? Yield strength of angle is 36 ksi and tensile strength is 58 ksi.



- (A) 68 kips
- (B) 81 kips
- (C) 85 kips
- (D) 135 kips

P0
053

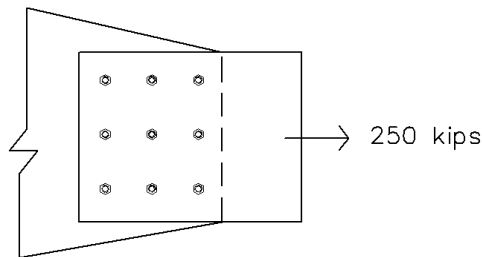
- 427.** For the L 5X3X1/2 connection below, what is the net effective area "Ae" for tensile rupture of the angle as defined by AISC specifications? The bolts used are 0.75 inch diameter. The longer leg of the angle is connected to the plate. Nominal Hole diameter = 13/16in



- (A) 2.72 in²
 (B) 2.98 in²
 (C) 3.32 in²
 (D) 3.37 in²

P0
054

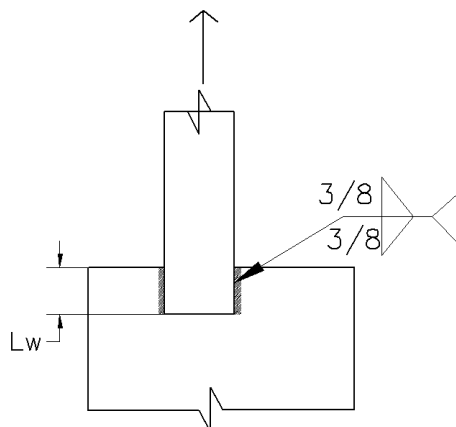
- 432.** Two plates are connected as shown in the plan below and are subjected to a service load of 250 kips. The thickness of the top plate is 5/8 in. The bolts are 3/4 inch in diameter and the holes are standard size. What is the bearing stress on the bolts holes?



- (A) 26 ksi
 (B) 32 ksi
 (C) 47 ksi
 (D) 55 ksi

P0
055

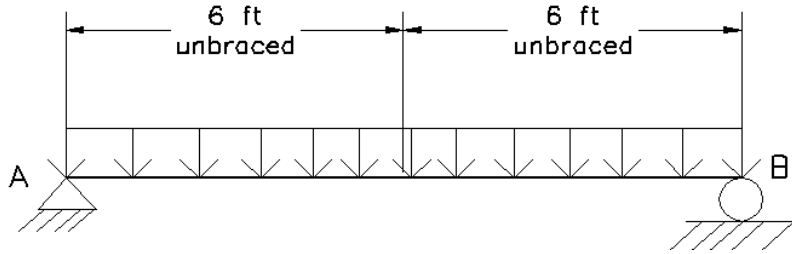
- 446.** A plate is connected to a gusset by welding on two sides as shown below. The service tension load acting concentrically on the welds are 10 kips dead load and 25 kips live load. Assuming that the base metal failure does not govern, what is the required length of weld "Lw" ?
 F_{exx} = 70 ksi



- (A) 2.75 in
 (B) 3.25 in
 (C) 3.5 in
 (D) 4 in

P0
056

- 341.** A simply supported 4X8 (nominal dimensions) southern pine section is subjected to a uniformly distributed load. The beam is braced laterally at midpoint. What is the beam slenderness ratio?



- (A) 7
(B) 8
(C) 9
(D) 10

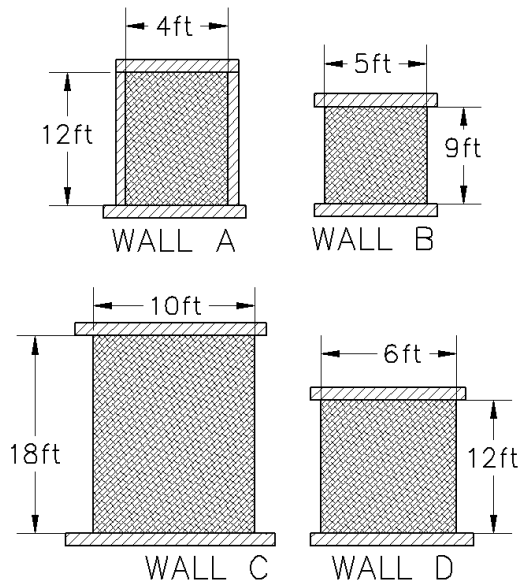
P0
057

- 375.** A single story building uses Southern Pine wood framed 3/8 in thick, blocked Structural -I WSP shear wall. 8d nails with spacing 2 inches are used at the panel edges. What is the allowable seismic shear capacity of the shear wall? Assume dry conditions.

- (A) 610 plf
(B) 1220 plf
(C) 1330 plf
(D) 1500 plf

P0
058

- 378.** Which of the following WSP shear walls does not comply with dimensional requirements of AWC- SDPWS provisions for wood framed shear wall?



- (A) Wall A
(B) Wall B
(C) Wall C
(D) Wall D

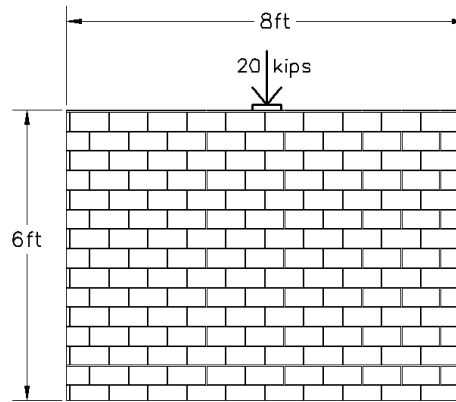
P0
059

- 303.** An unreinforced, fully grouted masonry wall spans 12 ft vertically from the slab on grade to the roof level. The load bearing wall is laterally supported by cross walls at 10 ft center to center. What is the minimum thickness of the wall to be considered for design?

- (A) 4 in
(B) 6 in
(C) 8 in
(D) 10 in

P0
060

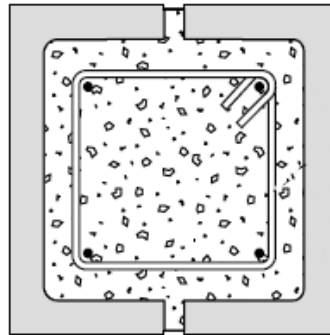
- 300.** What is the axial stress acting in the wall at mid height due to a 20 kips load acting on a 5 in X 5 in bearing plate on the top? The masonry units are standard hollow 8X8X16 CMU with face shell thickness of 1.25 in and web shell thickness of 1 in. Mortar is not provided at the webs of the units.



- (A) 133 psi
- (B) 157 psi
- (C) 195 psi
- (D) 250 psi

P0
061

- 306.** What is the minimum longitudinal reinforcement permitted in the masonry column shown below? The outer dimensions of the column are 9.625 in X 9.625 in.



- (A) Two #3
- (B) Two #4
- (C) Three #3
- (D) Four #3

P0
062

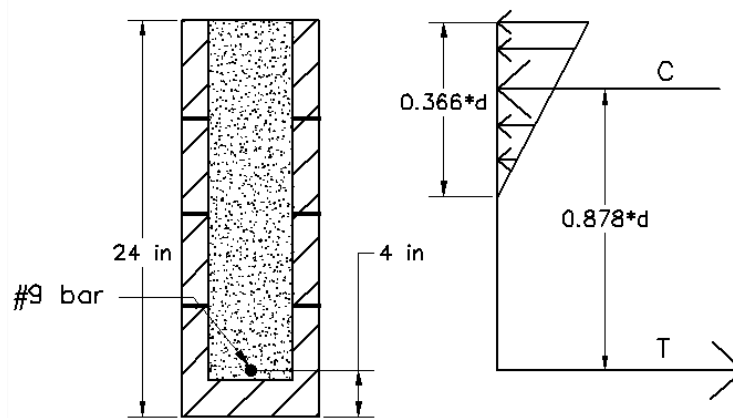
- 331.** What is the allowable bending moment for the reinforced masonry beam section below?
The section is flexurally reinforced with one no. 9 bar .

Section properties:

Rebar $f_y = 60$ ksi

Masonry $f'_m = 2000$ psi

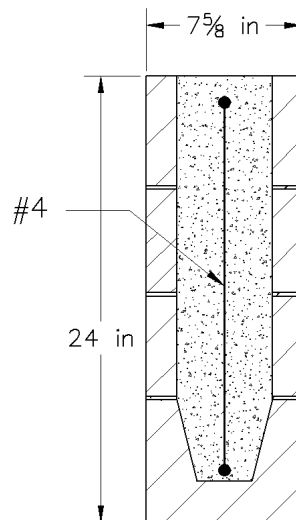
Width of the section = 7.625 in.



- (A) 20 kip*ft
(B) 27.5 kip*ft
(C) 32.25 kip*ft
(D) 36.75 kip*ft

P0
063

- 334.** For the masonry beam section shown below, shear stress resisted by masonry, $F_{vm} = 43.6$ psi and shear stress F_{vs} resisted by shear reinforcement = 27.5 psi . If a single legged #4 bar is used as shear reinforcement, what is the maximum spacing of the #4 bar permissible? Effective depth for tension bar, $d = 20$ in.



- (A) 6 in
(B) 9 in
(C) 15 in
(D) 24 in

P0
064

- 266.** 250 ft³ of concrete mix is to be produced. The mix proportion by weight is 1: 2 :3. Six gallons of water is to be used per bag of cement. How many bags of cement will be required for the job if the target air content in the mix is 5%?

Density of cement = 195 lbf/ft³

Density of fine aggregate = 160 lbf/ft³

Density of coarse aggregate = 164 lbf/ft³

Weight of cement bag = 94 lbs

- (A) 40 bags
- (B) 50 bags
- (C) 60 bags
- (D) 70 bags

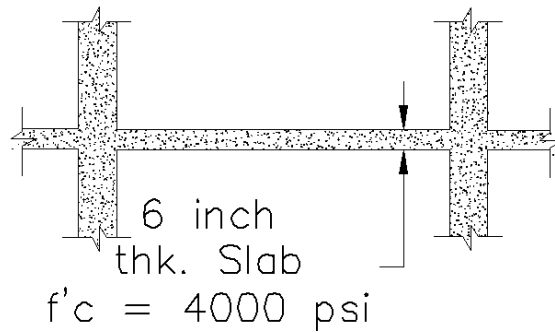
P0
065

- 215.** For a concrete construction job site, which of the following can be inspected intermittently and does not require constant monitoring by the special inspector?

- (A) Verifying proper spacing, cover and size of rebars,
- (B) Adhesive anchors installation to fasten electrical panel unit on a shear wall cast two weeks ago,
- (C) Verifying groove welds used for rebar lapping,
- (D) Application of prestressing forces on a multi span concrete girder.

P0
066

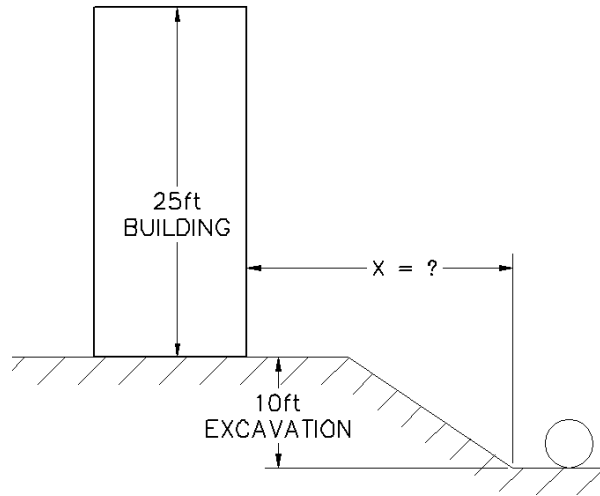
- 216.** For a 6 inch thick concrete slab shown below, #5 rebars are spaced at 3 inch c/c at the negative moment zones. What is the maximum aggregate size that can be approved by the special inspector for the pour?



- (A) 1.25 inch
- (B) 1.5 inch
- (C) 1.75 inch
- (D) 2 inch

P0
067

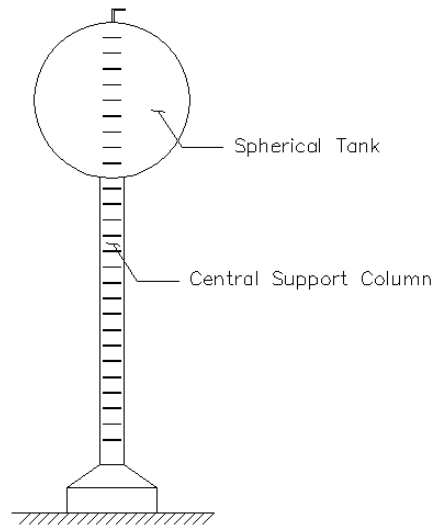
- 240.** Utility lines are to be laid next to a 25 feet tall building at a depth of 10 feet below grade. An undisturbed perimeter of at least half the height of the building has to be maintained around the building. What is the minimum horizontal distance “X” from the face of the building to the bottom of the excavated slope following OSHA’s excavation safety regulations? The soil around the building is Type C per OSHA classifications.



- (A) 18 ft
- (B) 22.5 ft
- (C) 27.5 ft
- (D) 30.75 ft

P1
068

- 1.** An elevated reinforced concrete spherical water tank can store 2000 gallons of water. The walls of the tank are 6 inches thick. What is the total unfactored dead load acting on the supporting column?



- (A) 17 kips
- (B) 24 kips
- (C) 30 kips
- (D) 34 kips

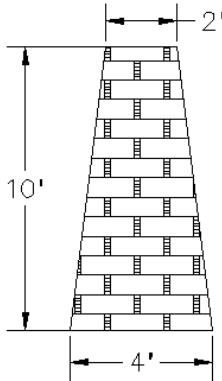
P1
069

- 19.** For a sloped roof, what is the basis for applying slope adjustment factors to reduce the live loads per the provisions of ASCE7-16 and IBC 2018?

- (A) Structural capacity of roofing members increases with inclination
- (B) Seismic load governs for sloped roof and live loads can be neglected
- (C) Sloped roofs have reduced accessibility and are less likely to be occupied
- (D) Snow load accumulation is lesser in sloped roofs

P1
070

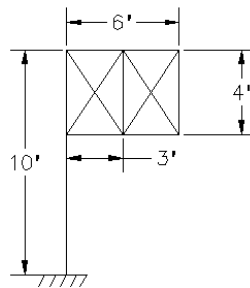
5. A trapezoidal stone masonry abutment is constructed to support a culvert. The abutment is 12 ft in length and the cross section of the abutment is as shown below. What is most nearly the self-weight of the abutment?



- (A) 35 kips
(B) 61 kips
(C) 76 kips
(D) 92 kips

P1
071

7. A cantilevered traffic light pole is designed as shown below. The vertical and horizontal members are fabricated with HSS 6X6X1/2 and the diagonal members are fabricated with HSS 2X2X1/4. The whole assembly is to be erected at the job site. What is the capacity of the hoisting mechanism required to lift the framing assembly? Assume additional rigging weight of 10% of the framing assembly.



- (A) 1165 lb
(B) 1282 lb
(C) 1382 lb
(D) 1400 lb

P1
072

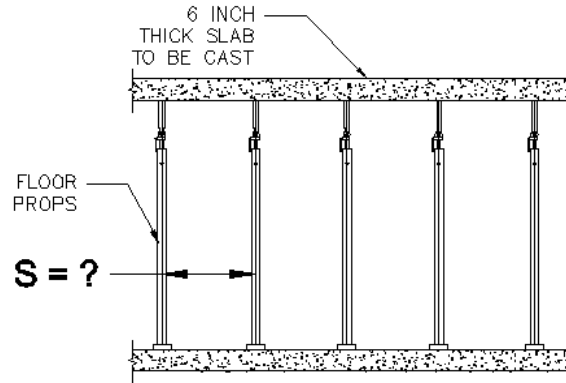
21. For a deep foundation construction job, H piles are used in support of excavation systems. The section to be used is HP 10X57. The yield stress of the material is 50 ksi. What is the allowable driving load for the top driven pile?



- (A) 500 kips
(B) 750 kips
(C) 1000 kips
(D) 1500 kips

P1
073

22. A 6-inch thick fresh concrete slab is to be cast on a new floor. The construction load is estimated as 50 psf including equipment and weight of formwork. The newly cast floor is to be shored with floor props that have allowable carrying capacity 2 kips. The props are to be laid out following a square layout, with equal spacing in both directions. What is the maximum permitted spacing? Unit weight of concrete = 150 pcf



- (A) 2ft
(B) 3ft
(C) 4ft
(D) 5ft

P1
074

29. An HVAC unit is to be installed at the roof of a 5 story building. The unit weighs 200 pounds. The unit is 6 ft tall. The length and width of the unit is 5ft each. Center of gravity of the unit can be assumed to be at the geometric center of the unit. Calculate the lateral wind load acting on the unit to be considered for anchorage design. The properties of the building are as follows:

Width of building = 40 ft Roof height = 50 ft
 $V = 95$ mph,
 Topographic factor $K_{zt} = 1$,
 Ground elevation factor $K_e = 1$

- (A) 0.15 kips
(B) 0.25 kips
(C) 0.5 kips
(D) 0.9 kips

P1
075

57. During an extreme rainfall event, for a roof in Miami, the hydraulic head for the design flow of the secondary drainage system is 2 inches. When the primary drainage system is blocked, the volume of ponding on the 30ftX30ft roof area is 5,000 gallons . What is the design rain load on the roof?

- (A) 13 psf
(B) 29psf
(C) 37psf
(D) 57psf

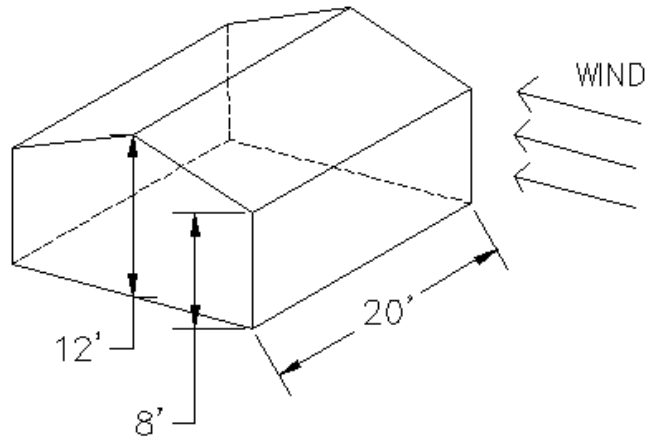
P1
076

62. The maximum moment acting on a single lane bridge girder due to truck load is 500 kip*ft. The maximum moment due to design lane load is 600 kip*ft. What is the total unfactored moment including any impact factor per AASHTO LRFD bridge design specifications?

- (A) 1100 kip*ft
(B) 1265 kip*ft
(C) 1385 kip*ft
(D) 1518 kip*ft

P1
077

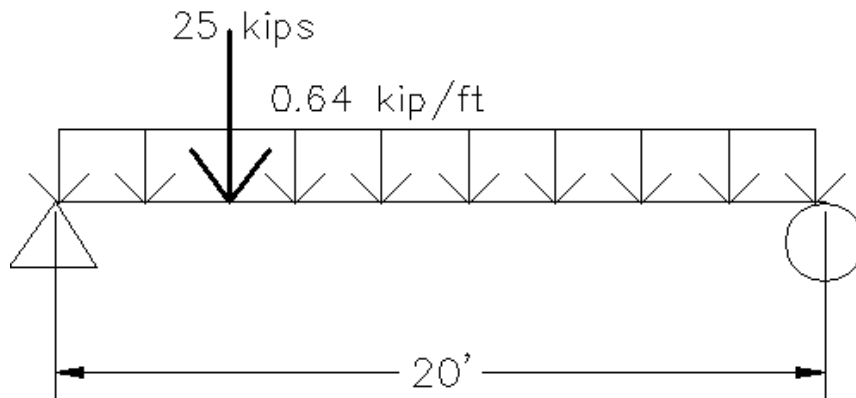
30. A regular shaped enclosed low rise building qualifies for wind analysis using envelope procedure. The calculations yielded a horizontal wind design pressure of 7 psf in the windward direction. What is the total wind load in the projected vertical plane of the windward side as shown below? Follow minimum design requirements per ASCE 7-16.



- (A) 1680 lb
(B) 3200 lb
(C) 3840 lb
(D) 4800 lb

P1
078

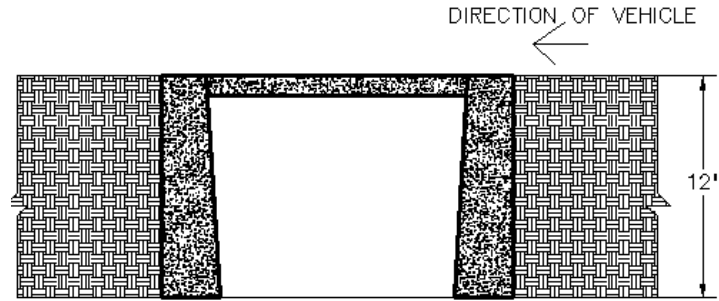
49. A single span simply supported bridge is subjected to a uniform lane load of 0.64 kip/ft and a moving concentrated wheel load of 25 kips. What is the maximum reaction acting on the support of the bridge?



- (A) 18.9 kips
(B) 25 kips
(C) 31.4 kips
(D) 37.8 kips

P1
079

- 70.** The height of an abutment wall supporting a bridge deck is 12ft. The abutment has backfill on one side as shown below. What is the additional lateral load due to vehicular load on top of the embankment? Calculate live load surcharge per AASHTO LRFD provisions.
- Active earth pressure coefficient = 0.33
 Earth pressure coefficient at rest = 0.5
 Passive earth pressure coefficient = 3
 Density of soil, $\gamma = 110$ pcf



- (A) 1288 lb/ft
 (B) 1848 lb/ft
 (C) 2048 lb/ft
 (D) 2288 lb/ft

P1
080

- 83.** Which of the following is not true about load combinations and associated limit states in AASHTO LRFD bridge design specifications?

- (A) For less important bridges, the load factors recommended by chapter 3 can be scaled down.
 (B) For design using extreme event combinations, inelastic behavior which may restrict operation is not permitted
 (C) Typically, 'Strength IV' combination governs over 'Strength I' if the dead load effects are at least two times the live load effects,
 (D) The design forces for jacking in service shall not be less than 1.3 times the permanent load reaction at the bearing, adjacent to the point of jacking

P1
081

- 84.** The load effects on the main girder of a single lane bridge are as follows:

- a. Service dead load moment = 800 kip*ft
 b. Service live load moment due to design truck = 250 kip*ft
 c. Service live load moment due to lane load = 300 kip*ft

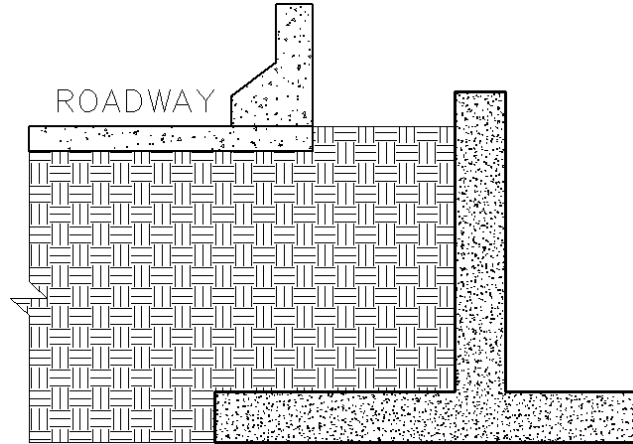
Force effects due to temperature, creep, settlement, water load and friction can be neglected,

What is the maximum factored moment acting on the girder per Strength I combination?

- (A) 2150 kip*ft
 (B) 2328 kip*ft
 (C) 2785 kip*ft
 (D) 2966 kip*ft

P1
082

- 85.** Partial section of an elevated roadway made on an embankment is shown below. The moment due to the back fill's unfactored active earth pressure on the wall is 10 kip*ft/ft. The moment due to live load surcharge from a vehicle travelling parallel to the wall is 5.5 kip*ft/ft. Moment due to wind load on the exposed stem of the wall above the backfill is 1.5 kip*ft/ft



What is the maximum factored moment acting on the wall per Strength V combination?

- (A) 20 kip*ft/ft
- (B) 22 kip*ft/ft
- (C) 24 kip*ft/ft
- (D) 26 kip*ft/ft

P1
083

- 87.** The cross-sectional areas of each of the shapes shown below are equal.



OPTION A



OPTION B



OPTION C



OPTION D

(X-SECTIONS ARE NOT TO SCALE)

Which shape is best suited for torsional applications?

- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

P1
084

- 110.** A cable is suspended between two points and is subjected to a uniformly distributed load of 200 lb/ft. The cable profile is parabolic. Which point in the cable is subjected to the lowest tension?

- (A) Support points
- (B) Mid span at lowest sag point
- (C) Quarter spans
- (D) Cannot be determined

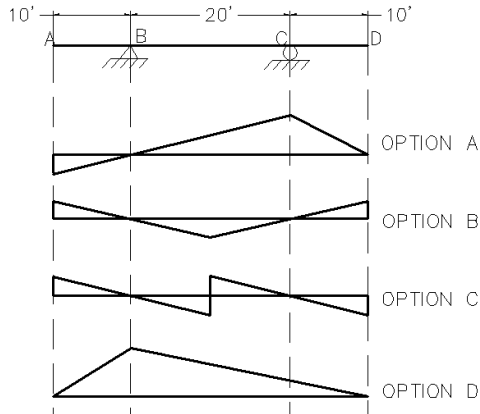
P1
085

- 170.** A solid circular shaft is subjected to a torsional moment of 10 ft-kips. The shaft is also subjected to a bending moment of 15 kip*ft. What is the equivalent twisting moment that when acting alone, produces the same maximum shearing stress as the combined effect of both moments acting currently?

- (A) 15 kip*ft
- (B) 18 kip*ft
- (C) 25 kip*ft
- (D) 28 kip*ft

P1
086

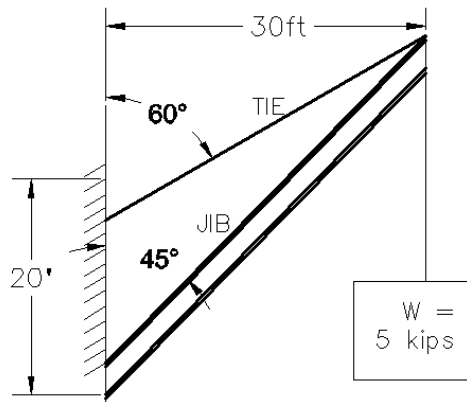
- 101.** For the beam shown below, select the correct influence line diagram for maximum shear at midspan, when a concentrated load passes from point A to point D.



- (A) Option A
(B) Option B
(C) Option C
(D) Option D

P1
087

- 111.** The configuration of a maintenance unit crane is as shown below. The pick weight is 5 kips. Both the tie and the jib are rigidly connected to a building wall. What is the axial force acting on the tie?



- (A) 8.66 kips
(B) 11.23 kips
(C) 13.66 kips
(D) 16.73 kips

P1
088

- 163.** Which of the following biaxial states of stress represents a body in pure shear?

(A)
$$\begin{bmatrix} \sigma_1 & \tau_{xy} \\ \tau_{yx} & \sigma_2 \end{bmatrix}$$

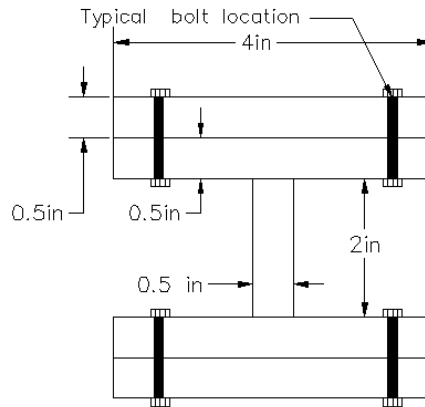
(B)
$$\begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}$$

(C)
$$\begin{bmatrix} \sigma_1 & 0 \\ 0 & -\sigma_1 \end{bmatrix}$$

- (D) None of the above

P1
089

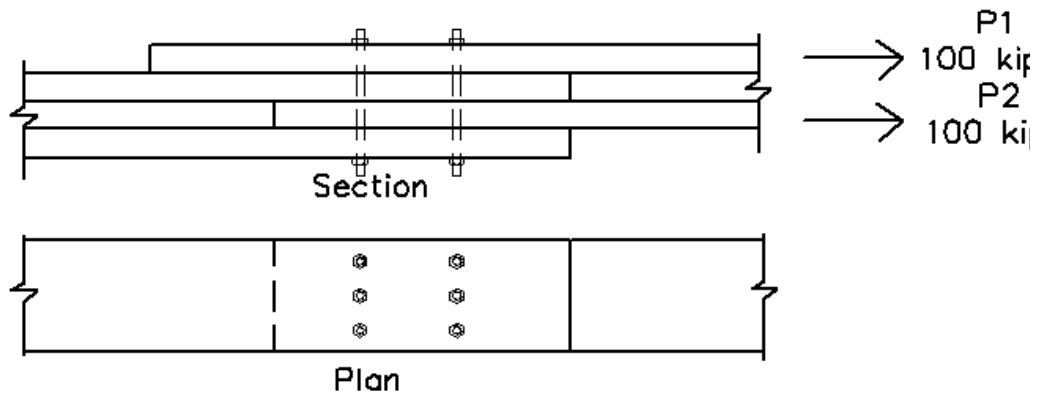
- 133.** A built up girder system is shown below. The maximum shear force acting in the section is 15 kips. The top and bottom flange plates are connected by 2 bolts on either side of the web as shown below. What is the maximum shear load on each bolt if the bolts are spaced at 12 inches center to center along the span?



- (A) 12.2 kips
(B) 16.5 kips
(C) 21.6 kips
(D) 24.8 kips

P1
090

- 134.** The figure below shows the section and plan of a 4 plate connection system. The top plate is subjected to a load of 100 kips and the 3rd plate is also subjected to a load of 100 kips. There are a total of 6 bolts as seen in the plan. What is the maximum nominal shear stress acting on the bolts? The bolts are 1 inch in diameter.



- (A) 21.23 ksi
(B) 42.46 ksi
(C) 63.69 ksi
(D) 84.92 ksi

P1
091

- 175.** A wooden block is fixed rigidly at the base. A tangential load is applied to the upper surface of the block. The deflection caused at the top of the block is due to

- (A) Shear and Bending moment,
(B) Bending moment only,
(C) Shear and torsional moment,
(D) Shear only,

P1
092

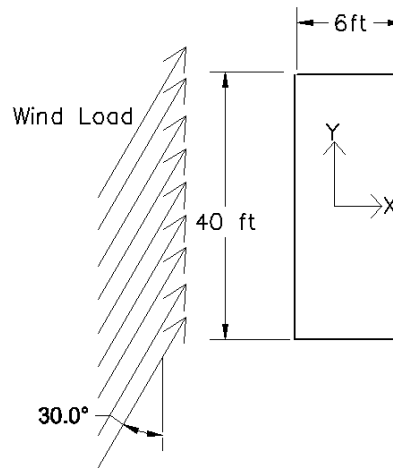
- 180.** A timber joist has a span 12 ft and is not supporting any ceiling. The joist is subjected to combined dead and live loading. What is the maximum allowable deflection for the joist?

- (A) 0.4 in
(B) 0.8 in
(C) 1 in
(D) 1.2 in

P1
093

- 142.** The plan of a 40 ft by 6 ft pile cap is shown below. The pile cap is 3 feet tall. It is supported by a pile that is 10 feet tall. What is the moment at the base of the pile about Y axis due to wind load of 50 psf acting at 30 degree inclination to the Y axis?

- (A) 21.7 kip*ft
(B) 32.8 kip*ft
(C) 33.5 kip*ft
(D) 43.3 kip*ft

P1
094

- 181.** A lintel supports unreinforced masonry above it. The span of the lintel is 8ft. What is the maximum allowable deflection in the lintel under combined service dead and live loads?

- (A) 0.16 in
(B) 0.26 in
(C) 0.53 in
(D) 1 in

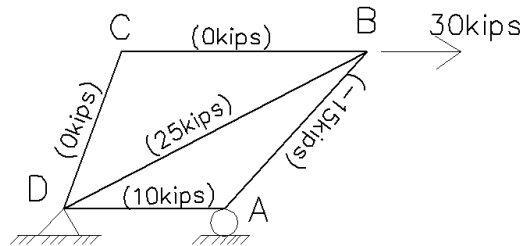
P1
095

- 184.** A simply supported truss is shown below. A 30 kips load acts at joint B. The forces resulting from the load on each member is shown on each member (tension +ve). What is the maximum horizontal deflection on joint B due to the load? Area = 10 in² and E = 29000 ksi for all members.

Lengths of members

- AB = 120 in
BC = 150 in
CD = 100 in
AD = 100 in
BD = 180 in

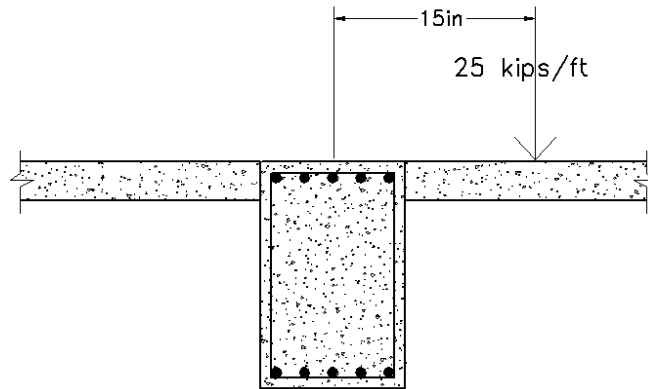
- (A) 0.011 in
(B) 0.017 in
(C) 0.023 in
(D) 0.027 in



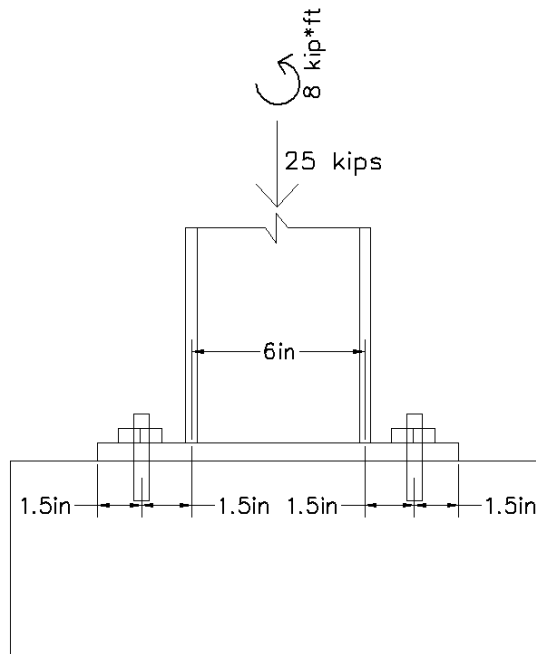
P1
096

- 195.** A reinforced concrete beam has a span of 20 ft. The beam is subjected to a live load of 25 kips/ft at a distance 15 in from the center of the beam as shown below. The beam is restrained against torsion only at one end. What is the torsional moment acting on the beam?

- (A) 125 kip*ft
(B) 250 kip*ft
(C) 312.5 kip*ft
(D) 625 kip*ft

P1
097

- 197.** The figure below shows a column base plate. The square base plate is 12 in X 12 in. Assuming the centroid of the compressive force in the base plate is at the flange of the column, what is the maximum bearing pressure exerted by the loads shown below?



- (A) 0.25 ksi
(B) 0.375 ksi
(C) 0.5 ksi
(D) 0.75 ksi

P1
098

- 202.** A building is to be analyzed for progressive collapse after sudden failure of ground floor column. Which of the following analysis is not required for alternate load path analysis?

- (A) Static analysis with factored service loads,
(B) Nonlinear analysis for large deformations,
(C) Dynamic analysis simulating sudden removal of column,
(D) Foundation uplift and bearing under failed column.

P1
099

238. Two adjacent buildings are to be constructed without setbacks from the property line. One of the buildings is 5 story tall and has a maximum inelastic response displacement of 7 inches per ASCE 7-16. Another building is 6 story tall and has a maximum inelastic response displacement of 9 inches. What is the required separation between the two buildings to prevent seismic pounding?

- (A) 9 in
- (B) 12 in
- (C) 16 in
- (D) 20 in

P1
100

392. A building frame consists of a combined dual wall and moment frame system to resist lateral loads. What is the primary purpose of the moment frame during a seismic event?

- (A) Transfer lateral loads directly to foundation,
- (B) Protect the shear walls from uncertainty in seismic loads,
- (C) Resist all the vertical loads,
- (D) Enhance lateral stiffness and resistance while the stiffer wall takes the majority of the lateral load.

P1
101

395. A thirty story building with a steel braced frame system is subjected to wind-induced vibrations. The building's natural frequency is close to the vortex shedding frequency of the wind. Which of the following is the most effective strategy to prevent structural damage due to large amplitude vibrations caused by resonance?

- (A) Increase the overall mass of the structure to counter wind loads and increase stability,
- (B) Increase the height of the building that leads to altered natural frequency of the building,
- (C) Install a tuned mass damper,
- (D) Reduce the stiffness of the braces to alter the natural frequency of the building.

P1
102

209. Which of the following is not true about procedures related to special inspections?

- (A) The owner, engineer of record, contractor and building officials have to approve the competency of the party serving as the special inspector,
- (B) For a welding job, in addition to non destructive evaluation reports, the qualifications of the fabricator also needs to be included as a part of inspection documentation,
- (C) Structural observations or special inspections are not mandatory for every project, regardless of the cost of the project,
- (D) The special inspector is not responsible for the contractor's means and methods.

P1
103

211. The statement of special inspections that is to be submitted to the building officials is prepared by

- (A) A firm hired by the owner to serve as the engineer of record of the project,
- (B) The contractor's team coordinating all construction permits ,
- (C) The special inspection agency hired for the project who is independent of the contractor,
- (D) The owner's technical personnel who oversee overall construction scheduling and official permits.

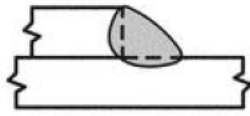
P1
104

218. Which document governs the content, format, overall procedures and approvals of construction submittals?

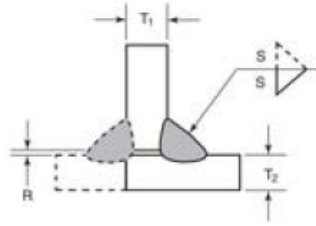
- (A) International Building Code ,
- (B) OSHA Technical manual,
- (C) ACI 318-14,
- (D) Contract documents and specifications.

P1
105

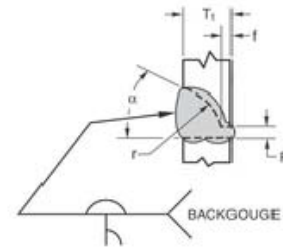
- 213.** Which of the following welds require special inspections for joint preparation prior to welding?



Option I



Option II



Option III

- (A) Option I
(B) Option II
(C) Option III
(D) Option I and II

P1
106

- 220.** Which of the following is typically not a part of the general contractor's submittal package for a commercial building project?

- (A) Vendor provided flooring product data sheet to be used in the building,
(B) Concrete cylinder break reports provided by the testing lab,
(C) Structural calculations for construction loads,
(D) Mill test reports.

P1
107

- 225.** Guy ropes are used to stabilize a bridge falsework system. The maximum pretension in the guy rope is 2.5 kips. The guy rope is inclined at 45 degrees to the horizontal and is anchored to a 4 ksi concrete slab. The table below shows ultimate forces that the anchors can withstand in lbs. A factor of safety of 2 can be used to convert the ultimate forces to allowable forces. What is the minimum diameter of the anchor that can be used?

Anchor Dia. (in)	Required Torque (ft-lb)	Embedment depth (in)	$F'_c = 2000 \text{ psi}$		$F'_c = 4000 \text{ psi}$	
			Tension	Shear	Tension	Shear
$\frac{1}{4}$	8	$2\frac{1}{8}$	2,260	1,680	3,300	1,680
$\frac{3}{8}$	25	4	4,800	4,000	5,940	4,140
$\frac{1}{2}$	55	6	5,340	7,240	9,640	7,240
$\frac{5}{8}$	90	$7\frac{1}{2}$	7,060	9,600	15,020	11,900

- (A) 1/4 in
(B) 3/8 in
(C) 1/2 in
(D) 5/8 in

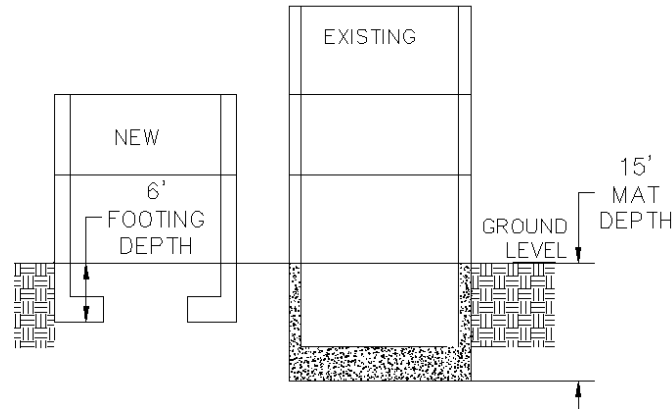
P1
108

- 234.** A multi level guard rail is installed on a single point adjustable suspension scaffold. What is the design moment acting on the midrail per OSHA regulations, if it is welded between two posts spaced 5 ft apart.

- (A) 50 lbs*ft
(B) 75 lbs*ft
(C) 100 lbs*ft
(D) 150 lbs*ft

P1
109

- 236.** A two story building is to be constructed next to an existing building with a basement. The depth of foundation of the new building is 6 feet and the depth of basement mat of the existing building is 15 feet below grade. Assuming that the footing load due to the new structure follows a 45 degree influence line, what is the minimum distance between the two buildings so that the existing building is away from the influence of new building loads?



- (A) 6 ft
- (B) 8 ft
- (C) 9 ft
- (D) 15 ft

P1
110

- 251.** Standard penetration test was conducted at a site. At the required depth, the effective overburden pressure was 4 ksf and the energy corrected N_{60} value was 20 blows. What is the estimated in situ dry unit weight of the soil sample?

Data obtained from lab tests are:

$e_{\max} = 0.4$ $e_{\min} = 0.2$ specific gravity of soil sample, $G_s = 2.7$

- (A) 115.2 pcf
- (B) 120.8 pcf
- (C) 125.4 pcf
- (D) 131.6 pcf

P1
111

- 263.** A concrete mix is designed for a roof slab that will be subjected to snow and ice accumulation and frequent rain. The nominal aggregate size is 3/4 inch and the compressive strength of concrete is 4500 psi. What is the recommended water - cement ratio and total air content in the concrete?

- (A) Target air content 6 % and minimum w/c ratio 0.45
- (B) Target air content 5 % and minimum w/c ratio 0.55
- (C) Target air content 6 % and maximum w/c ratio 0.45
- (D) Target air content 5 % and maximum w/c ratio 0.55

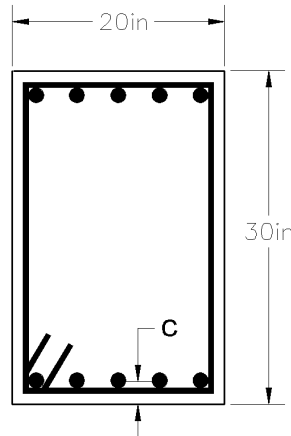
P1
112

- 278.** Which of the following is true about effects of temperature on different properties of steel?

- (A) Increase in temperature increases the yield and ultimate strength of steel.
- (B) Increase of temperature reduces the ductility and hardness of steel.
- (C) Temperature has no effect on the modulus of commonly used structural steel.
- (D) At lower temperatures, the stiffness of steel members increases.

P1
113

- 265.** The section of a precast reinforced concrete beam that is manufactured under controlled conditions is shown below. The beam is a part of an open parking lot in a rainy area. What is the minimum distance “c” shown in the figure below? The main bars are number 8 bars and the stirrups are number 4 bars.



- (A) 2 in
(B) 2.5 in
(C) 3 in
(D) 3.5 in

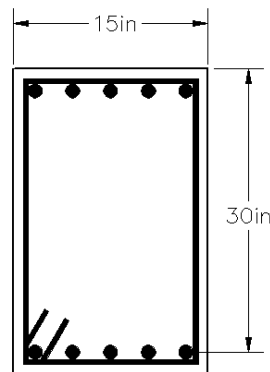
P1
114

- 287.** Which of the following is not true about the properties of cold formed steel ?

- (A) Cold formed steel exhibits anisotropic material behavior due to the forming process,
(B) Cold formed steel sections develop longitudinal residual stresses due to differential plastic deformation during forming,
(C) Cold forming process induces plastic deformation, particularly at the corners, leading to strain hardening which leads to increased yield strength in these regions,
(D) Because of the difficulty of cold working instead of being hot rolled, the dimensional tolerances for cold formed steel are higher than that for hot rolled sections.

P1
115

- 308.** A reinforced concrete beam is designed and approved to be constructed as shown in the section below. What is the minimum tolerable dimension of the effective depth ‘d’ while placing the rebars at the construction site per ACI 318-14?



- (A) 29 in
(B) 29.375 in
(C) 29.5 in
(D) 29.75 in

P1
116

- 310.** Two rebars shown below are identified as ASTM A615 and ASTM A706. Which of the following is true about the rebars?



- (A) Rebar with marking C denotes carbon steel Rebar
- (B) Rebar with marking S denotes ASTM A706 low-alloy steel rebar.
- (C) ASTM A615 rebars are more suitable for seismic reinforcement than A706 rebars because they are stronger than A706 rebars.
- (D) ASTM A615 rebars are more affordable and less ductile than A706 rebars.

P1
117

- 313.** After a standard penetration test, which of the following properties of the soil can be correlated with the corrected number of blows 'N' ?
- I) Unconfined compressive strength
 - II) Relative density
 - III) Angle of internal Friction

- (A) I and II
- (B) II only
- (C) I, II and III
- (D) I only

P1
118

- 269.** What is the cracked moment of inertia of a singly reinforced rectangular section with depth 24 in and width 12 in?. The beam is reinforced with 4#6 bars at the bottom and the effective depth is 22 in. The modulus of elasticity of concrete is 3600 ksi and the modulus of elasticity of rebar is 29000 ksi.

- (A) 924 in⁴
- (B) 1875 in⁴
- (C) 4494 in⁴
- (D) 4804 in⁴

P1
119

- 275.** The gross area of a prestressed girder is 500 in². The profile of tendons are concentric with the gross area of the girder throughout the span and the jacking force is 350 kips. The concrete is normal weight concrete and the strength of concrete during initial prestressing is 5000 psi. What is the estimated loss in prestressing because of shortening of concrete?

- (A) 3.75 ksi
- (B) 4.46 ksi
- (C) 5.45 ksi
- (D) 6.51 ksi

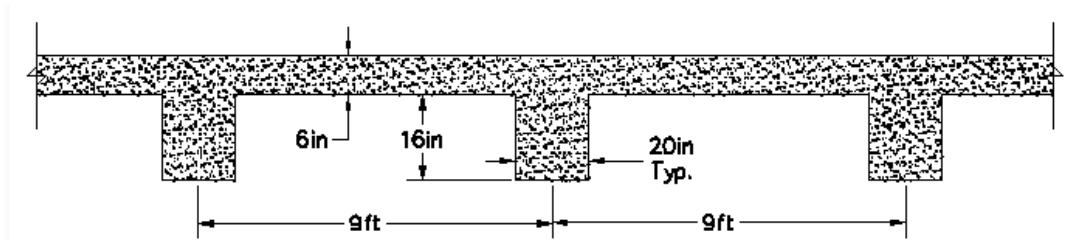
P1
120

- 346.** A four span continuous one way slab is constructed using Gr 60 rebar and lightweight concrete with $f'_c = 4000$ psi and unit weight 110 pcf. The interior spans are 21 ft in length and exterior spans are 17 ft in length. What is the minimum required thickness of the slab?

- (A) 6 in
- (B) 8.5 in
- (C) 9 in
- (D) 10 in

P1
121

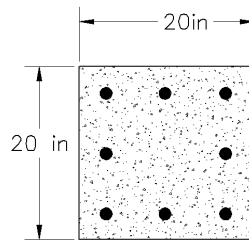
- 348.** What is the effective T- beam flange width for the center beam shown below ? The span of the RCC beams are 30ft and the clear span is 27ft.



- (A) 48 in
(B) 96 in
(C) 101 in
(D) 108 in

P1
122

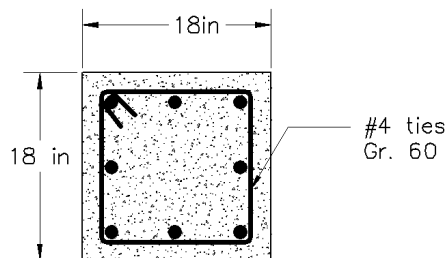
- 384.** What is the design shear strength of the column section shown below? The column is subjected to a factored axial load of 150 kips.
 $f'_c = 4000$ psi $f_y = 60$ ksi



- (A) 25 kips
(B) 38.3 kips
(C) 45.1 kips
(D) 52.5 kips

P1
123

- 385.** The column shown below is subjected to a factored shear of 75 kips. What is the required tie spacing to satisfy the strength requirement. Assume that the entire shear is taken by the reinforcement only. Effective cover for longitudinal bars = 3 inches.



- (A) 3.5 in
(B) 4.75 in
(C) 6 in
(D) 9 in

P1
124

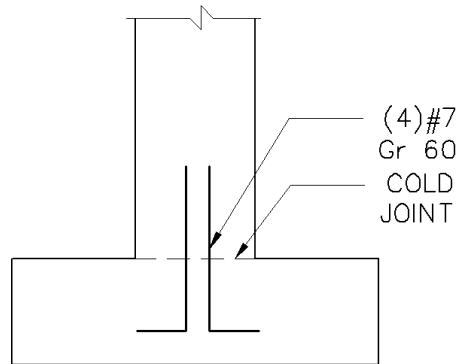
- 301.** Which of the following is not true about the property of mortar used in masonry construction?

- (A) During construction, the temperature of the mortar mix should always be maintained less than 120°F and above 40°F.
(B) The strength of the mortar should always be greater than the strength of the individual masonry units,
(C) Type M mortars have high strength and is used for heavy duty applications,
(D) Type N and Type O mortars have high workability.

P1
125

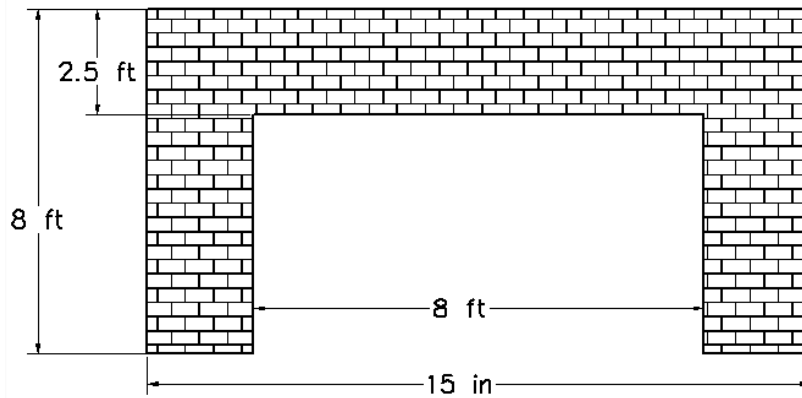
- 389.** A 20in X 20in column is cast over an existing footing as shown below. Four #7 dowels connecting the column and the footing are Gr. 60 rebar. The compressive strength of the footing's concrete is 3 ksi and the compressive strength of the column's concrete is 4 ksi. What is the design capacity of the column base in shear friction?
 $\Phi = 0.75$, Area of #7 bar = 0.6 in^2

- (A) 65 kips
 (B) 86 kips
 (C) 108 kips
 (D) 140 kips



P1
126

- 335.** What is the effective span of the masonry beam shown below? Beam is not built integrally with the piers.



- (A) 8 ft
 (B) 10.5 ft
 (C) 11.5 ft
 (D) 15 ft

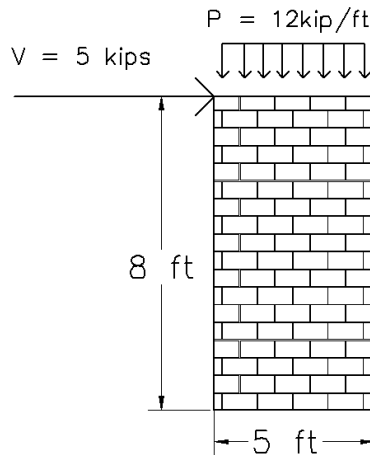
P1
127

- 314.** A concrete masonry assembly is constructed using individual units of strength 2000 psi tested per ASTM C90 and using Type N mortar. Thickness of mortar beds is $\frac{1}{2}$ inch. What is the specified compressive strength 'f'm' of the masonry assembly?

- (A) 1500 psi
 (B) 1750 psi
 (C) 2000 psi
 (D) 2650 psi

P1
128

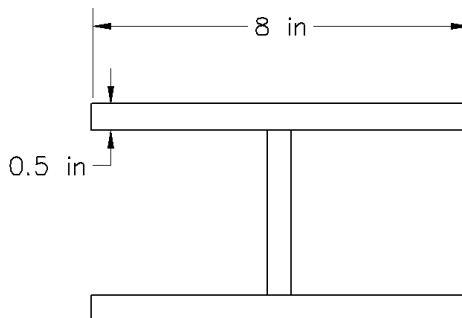
- 373.** A fully grouted masonry wall is subjected to a 12 kip/ft distributed load at the top and a shear of 5 kips acting at the top. The thickness of the wall is 7.625 in and the total area of reinforcement on the both ends is 4 in². The effective cover for the rebar is 4 in. What is the maximum tensile stress acting on the section?



- (A) 0
(B) 0.131 ksi
(C) 0.105 ksi
(D) 0.15 ksi

P1
129

- 321.** A symmetric plate girder is constructed using A 572 Gr. 50 plates. The section of the girder is shown below. Based on the flange dimensions, which of the following is true about the compactness of the flange if the girder is subjected to axial loads and moments?



- (A) Compact
(B) Non Compact
(C) Slender
(D) Cannot be determined with given information

P1
130

- 370.** The effective length of a W14X82 Grade 50 column is 15 ft in both directions. What is the maximum allowable axial compression it can carry, if it is also subjected to a strong axis flexure of 80 kip*ft and weak axis flexure of 40 kip*ft. Lateral torsional buckling is prevented by bracing the compression flanges only.

$$E = 29000 \text{ ksi}$$

- (A) 125 kips
(B) 201 kips
(C) 233 kips
(D) 489 kips

P1
131

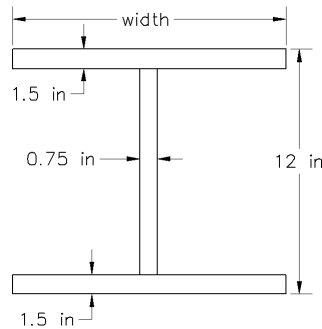
- 326.** Standard wide flange sections are used as horizontal members in a steel construction project to resist transverse loads. Transverse stiffeners are not used in the S beams. The analysis results show that maximum service level shear acting in the beam is 46 kips. What is the most economical section that can be used in the project? All beams have sufficient axial and flexural strengths.



- (A) S6X17.25
- (B) S8X18.4
- (C) S8X23
- (D) S10X25.4

P1
132

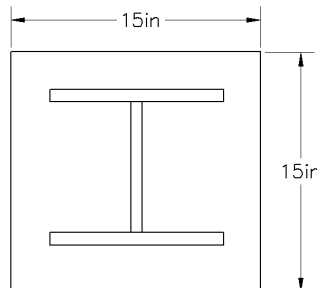
- 360.** For the built up column shown below, subjected to axial compression only, what is the maximum flange width permissible, so that the flanges are non slender?
 $F_y = 50$ ksi, $E = 29000$ ksi



- (A) 20 in
- (B) 28 in
- (C) 35 in
- (D) 40 in

P1
133

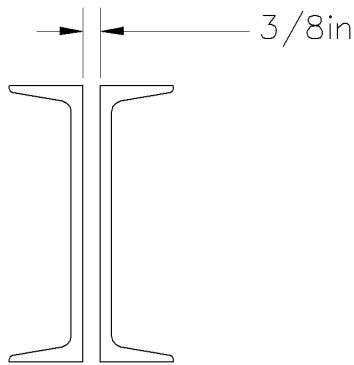
- 362.** A W section is centered on a 15 in X 15 in base plate as shown below. The overall depth of the section, $d = 11$ in and flange width, $b_f = 12$ in. What is the minimum required thickness of the base plate to support a factored axial load of 650 kips? $F_y = 50$ ksi



- (A) 1 in
- (B) 1.125 in
- (C) 1.375 in
- (D) 1.625 in

P1
134

- 365.** A built up column section is made with two channels C 12X30 spaced at $3/8$ in back to back as shown below. The effective length in both X and Y direction is 10 ft. What is the nominal critical buckling stress for the column? $F_y = 36$ ksi, $E = 29000$ ksi



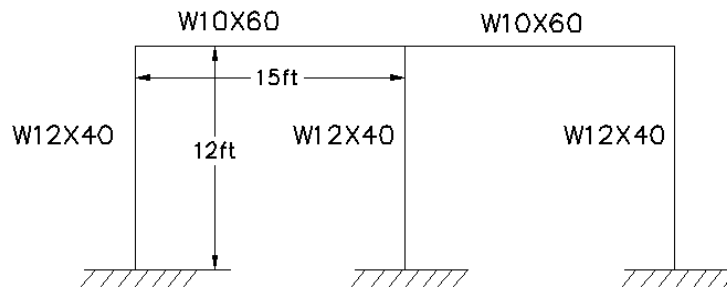
- (A) 15 ksi
- (B) 20 ksi
- (C) 25 ksi
- (D) 30 ksi

P1
135

- 368.** For the moment resisting frame below, what is the effective length factor for the leftmost column?

$E = 29000$ ksi

$F_y = 50$ ksi



- (A) 1
- (B) 1.2
- (C) 1.3
- (D) 2

P1
136

- 279.** The yield strength of the threaded rod shown below is 120 ksi. The minimum ultimate strength is 150 ksi. What is the minimum allowable tensile strength and the minimum allowable shear strength for bearing type connections using the rod if the material conforms to AISC specifications?



- (A) 56 ksi and 33 ksi
- (B) 84 ksi and 50.625 ksi
- (C) 90 ksi and 54 ksi
- (D) 112.5 ksi and 67.5 ksi

P1
137

280. An ASTM A307 bolt is subjected to a shear stress of 10 ksi. What is the reduced allowable tensile stress in the bolt material when subjected to shear tension interaction?

- (A) 8 ksi
- (B) 12 ksi
- (C) 25 ksi
- (D) 35 ksi

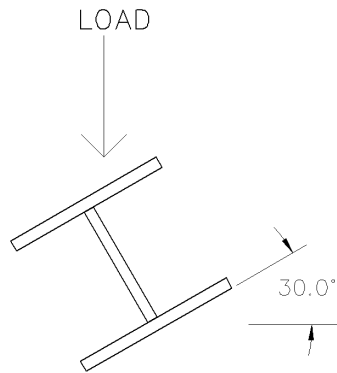
P1
138

290. A threaded rod is pretensioned to 102 kips during installation. The slip critical connection is in double shear and the surface can be classified as class A. The bolt holes are standard size. What is the allowable shear strength of the connection? There is only one shim plate between the connected parts.

- (A) 20 kips
- (B) 22 kips
- (C) 40 kips
- (D) 44 kips

P1
139

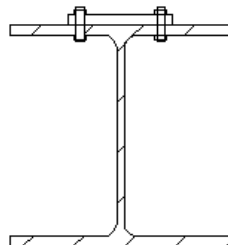
323. A simply supported roof purlin has a span of 12 ft. The uniformly distributed load acting on the purlin is 1 kip/ft. If the allowable flexural strengths for bending in strong axis and weak axis are 30 kip*ft and 10 kip*ft respectively, what is the total flexural utilization of the section?



- (A) 50%
- (B) 75%
- (C) 90%
- (D) 150%

P1
140

418. A 3/4 inch thick plate is to be bolted on top of a W section flange as shown below using standard size holes. The diameter of the bolt is 1/2 inches. What is the minimum clear distance from the edge of the plate to the edge of the hole?

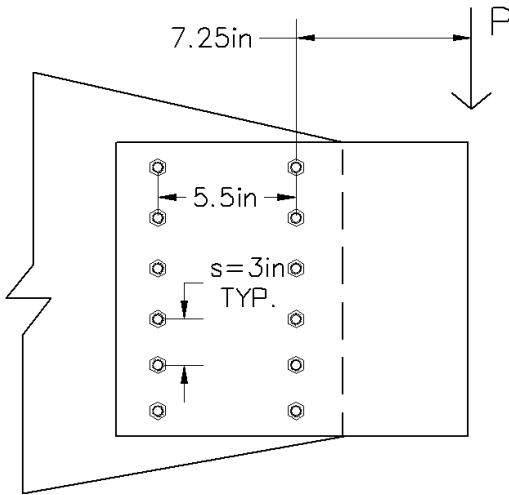


- (A) 15/32 in
- (B) 9/16 in
- (C) 5/8 in
- (D) 3/4 in

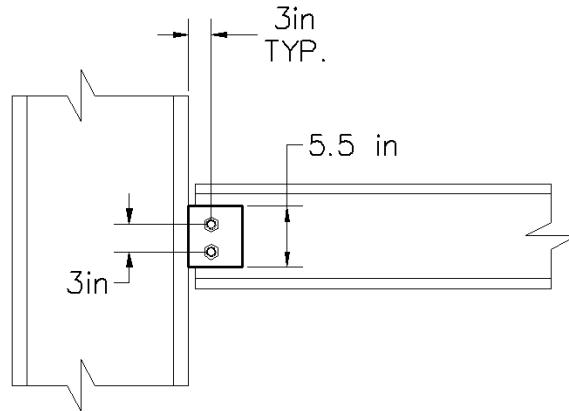
P1
141

- 422.** Using the instantaneous center of rotation method, what is the maximum allowable load “P” that the bolted connection shown below can withstand? The governing failure of the connection is bolt shear strength and allowable shear strength of each bolt is 20 kips.

- (A) 55 kips
(B) 78 kips
(C) 105 kips
(D) 200 kips

P1
142

- 434.** What is the allowable vertical reaction that can be transferred by the single plate shear connection shown below? The 3/8 inch thick shear plate is sufficiently welded to the column and has sufficient edge distances per AISC requirements. The bolts used are 3/4 inch diameter A325 and threads are not included in the shear plane. Plate $F_y = 36$ ksi.



- (A) 21.1 kips
(B) 25.3 kips
(C) 27 kips
(D) 29 kips

P1
143

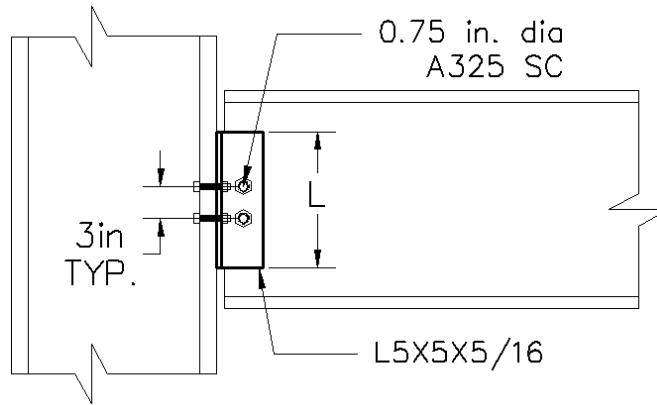
- 436.** For a beam to column moment connection, which of the following factors most significantly affects the rotational stiffness of the connection?

- (A) The grade of steel used for beams and columns,
(B) The length of the fillet welds along the beam web,
(C) The diameter of bolts used in the flange connection plates,
(D) The thickness and size of flange plates or angles,

P1
144

- 443.** A double angle beam to column shear connection is made using 5X5X5/16 angles on both sides of the beam web as shown below (number of bolts shown for illustration of connection type only). The spacing of the bolts will be kept at 3 inches consistent and Length of the angle 'L' varies per number of bolts used such that the top and bottom bolt edge distances are at least 1.25 in. The horizontal edge distance for the bolts will be maintained at 1.375 in. minimum. The connection has to transfer a service dead load reaction of 20 kips and a service live load reaction of 50 kips. Assuming failure of connection is governed by the angle and bolts, what is the number of slip critical (class A) A325 , 3/4 inch diameter bolts required for the beam web? Beams and columns are A992. Yield strength of angle = 36 ksi
Tensile strength = 58 ksi.
All holes are standard size.

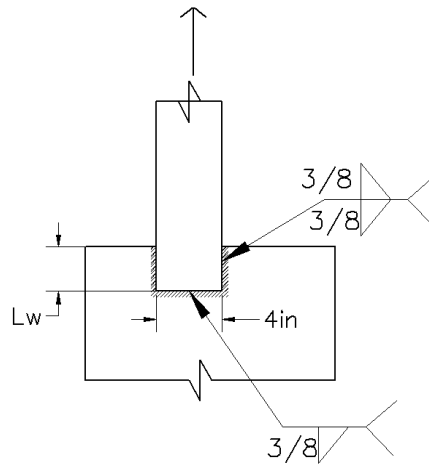
- (A) 2
(B) 4
(C) 6
(D) 8



P1
145

- 447.** A plate is connected to a gusset by welding on two sides as shown below. The service tension load acting concentrically on the welds are 20 kips dead load and 30 kips live load. Assuming that the base metal failure does not govern, what is the required length of weld "Lw" ?
F_{exx} = 70 ksi

- (A) 1.5 in
(B) 2 in
(C) 2.25 in
(D) 3 in



P1
146

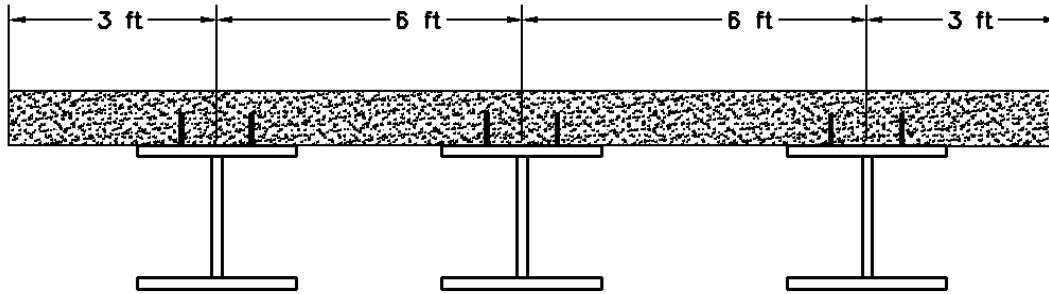
- 414.** A composite steel deck with 5 inch thick concrete slab and Gr. 50-W18X71 beams, designed for full composite action is unshored before pouring. The factored dead load moment including weight of slab and beam is 75 kip*ft and factored live load moment is 120 kip*ft. What is the factored bending stress at the bottom fibre of the steel beam immediately after pouring?

Section modulus of transformed section measured to the bottom of the section = 185 in³

- (A) 4.86 ksi
(B) 7.1 ksi.
(C) 10.3 ksi
(D) 13.4 ksi

P1
147

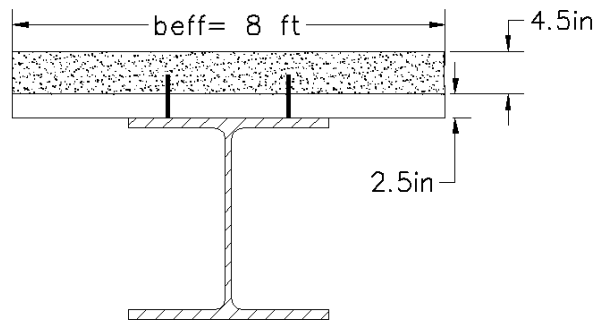
- 396.** The section of a composite deck system is shown below. What is the total effective width of the concrete slab to be considered for the composite beam on the edge? The center to center distance between supports of the beam along the span is 16ft.



- (A) 2ft
(B) 4ft
(C) 6ft
(D) 8ft

P1
148

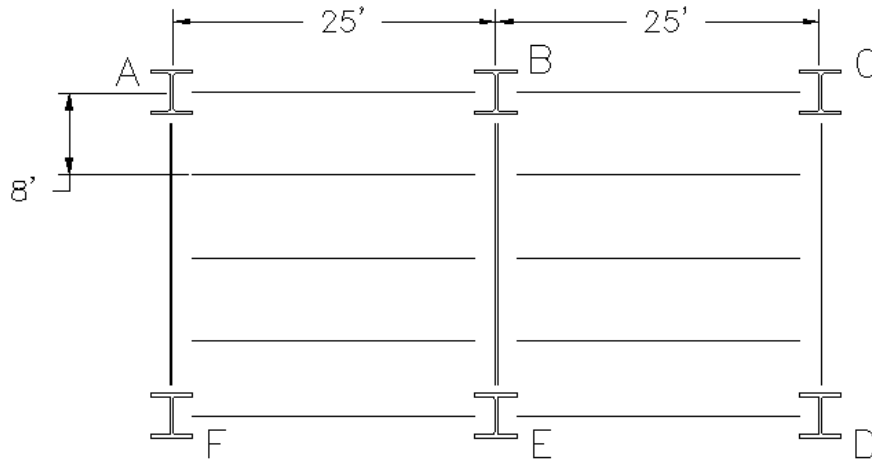
- 401.** A composite steel beam with W18X50 steel section has the steel decking ribs perpendicular to the span of the beam as shown below. The modular ratio of steel to concrete is 8. What is the location of the elastic neutral axis (strong axis) of the composite member from the top of the composite section?



- (A) 5.19 in
(B) 6.23 in
(C) 11.32 in
(D) 16.11 in

P1
149

- 413.** Partial framing plan of a building is shown below. The spandrel beam AB supports a 13 ft tall, 8 in thick wall with unit weight 80 lbf/ft². The floor dead load and live loads are 75 lbf/ft² and 100 lbf/ft². Assuming simple connections and full lateral support of compression flange, what is the most economical section that satisfies strength requirements for beam AB? Material used is Gr. 50.



- (A) W12X30
- (B) W12X40
- (C) W12X45
- (D) W12X53

P2
150

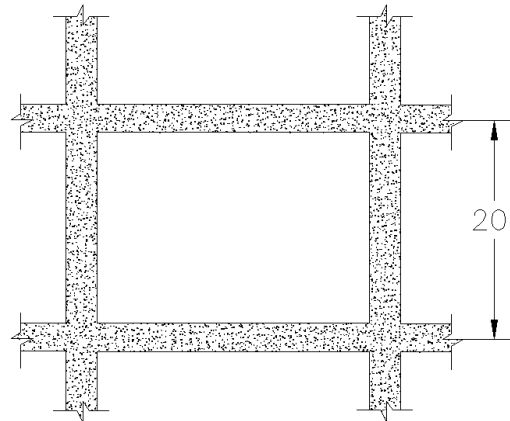
- 127.** The frame below shows partial elevation of beams and columns in an intermediate moment resisting, reinforced concrete system. The center to center vertical distance between the beams i.e. center to center height of columns is 20ft. The depth of both top and bottom beams is 24 inches. The nominal flexural strength at the top and bottom of the columns is 450 kip*ft and 600 kip*ft. The unfactored shears acting on the columns are as follows:

Dead load shear = 5 kips

Live load shear = 10 kips

Earthquake load shear = 20 kips

What is the factored design shear to be considered for the column?



- (A) 45 kips
- (B) 60 kips
- (C) 76 kips
- (D) 89 kips

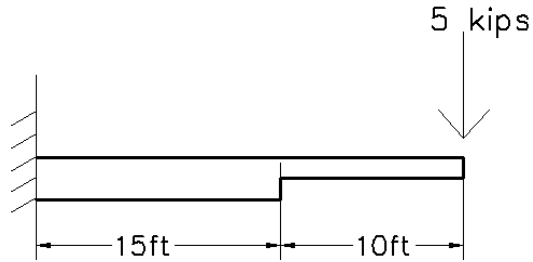
P2
151

- 307.** What is the fire resistance rating of a concrete masonry wall constructed of calcareous gravel that has an equivalent thickness of 5 inches?

- (A) 1.5 hour
- (B) 2.25 hour
- (C) 2.75 hour
- (D) 3 hour

P2
152

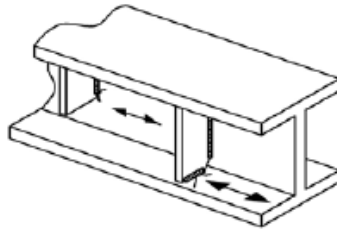
- 179.** A cantilevered beam is loaded at the tip as shown below. The heavier first 15 ft length of the beam has a moment of inertia 1000 in^4 and the lighter 10ft length has a moment of inertia of 500 in^4 . What is the deflection at the tip of the beam? $E = 29000 \text{ ksi}$



- (A) 1 in
(B) 1.35 in
(C) 1.65 in
(D) 2 in

P2
153

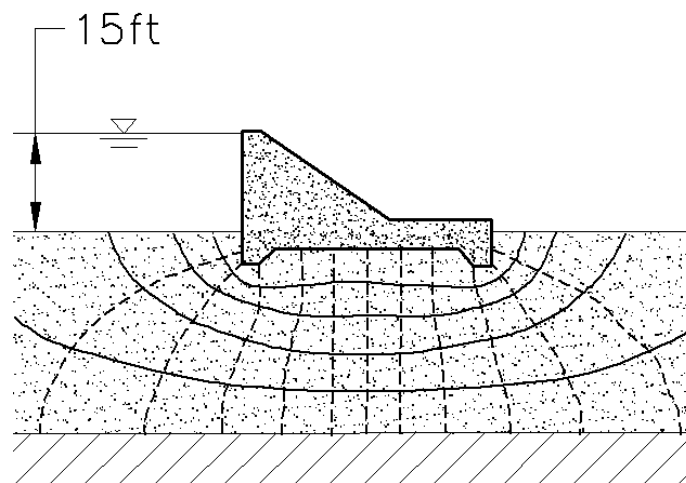
- 207.** A gusset plate is welded to the web of a bridge girder as shown below, near an interior support. The span of the girder is 60 ft and is a part of a continuous girder system. Calculate the minimum fatigue resistance of the welded plate. The single lane average daily truck traffic is 2000 vehicles per day.



- (A) 3.77 ksi
(B) 4.21 ksi
(C) 5.38 ksi
(D) 7.63 ksi

P2
154

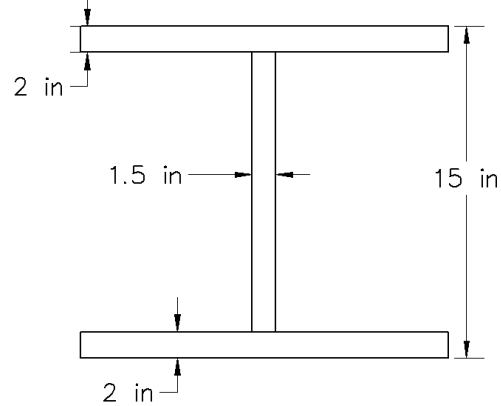
- 258.** The figure below shows a flow net for seepage under an embankment. Under the flow net, there is an impervious layer. The permeability of the soil is 0.0001 ft/min . What is the estimated seepage per day?



- (A) 0.9 cubic feet/ day/feet
(B) 1.1 cubic feet/ day/feet
(C) 1.3 cubic feet/ day/feet
(D) 1.5 cubic feet/ day/feet

P2
155

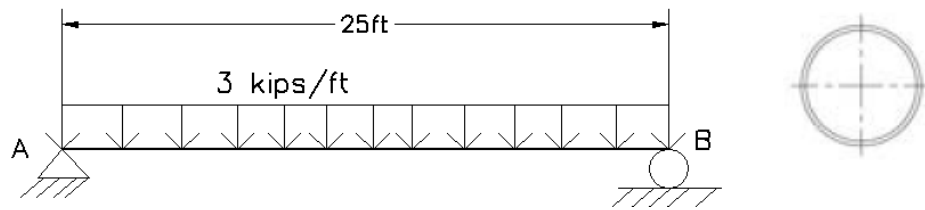
- 288.** A built up plate girder is made using A572 Gr 50 steel plates. The webs of the girder are stiffened with transverse stiffeners with 40 inches clear spacing between the stiffeners. What is the web plate shear buckling coefficient per AISC specifications?



- (A) 5
(B) 5.34
(C) 5.53
(D) 5.7

P2
156

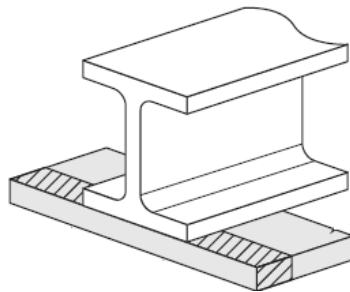
- 320.** A simply supported steel beam has a span of 25 ft and is braced at the midpoint. The beam is subjected to a uniformly distributed live load of 3 kip/ft. What is the lightest ASTM A500 Gr. C, 46 round HSS section that will satisfy the flexural strength requirements?



- (A) HSS 16.000 X 0.375
(B) HSS 16.000 X 0.438
(C) HSS 18.000 X 0.375
(D) HSS 18.000 X 0.5

P2
157

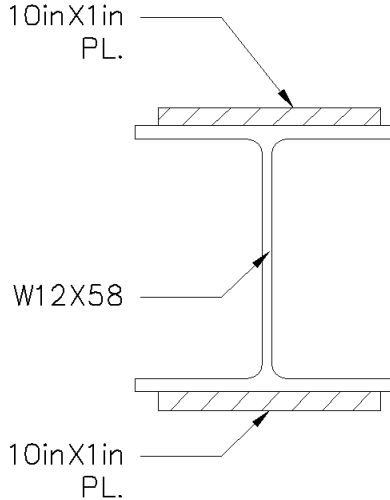
- 322.** A simply supported Gr. 50 W30X90 beam is subjected to a uniformly distributed factored load of 10 kip/ft throughout the span of the beam. The span of the beam is 25 ft. What is the minimum length of the bearing plate 'lb' along the span of the beam?



- (A) 2.5 in
(B) 3.5 in
(C) 4.5 in
(D) 5.5 in

P2
158

- 324.** 10 inch X 1 inch cover plates are welded effectively to the top and bottom flanges of a W12X58 beam section as shown below. What is the nominal plastic moment of resistance of the section about the major axis? $F_y = 50$ ksi for both plates and rolled section.



- (A) 850 kip*ft
(B) 910 kip*ft
(C) 980 kip*ft
(D) 1055 kip*ft

P2
159

- 328.** What is the maximum allowable concentrated load that can be applied to a Gr 50 ksi W10X30 section without the use of web stiffener plates. Assume load is applied sufficiently away from supports and length of concentrated load along the span = 1 in.

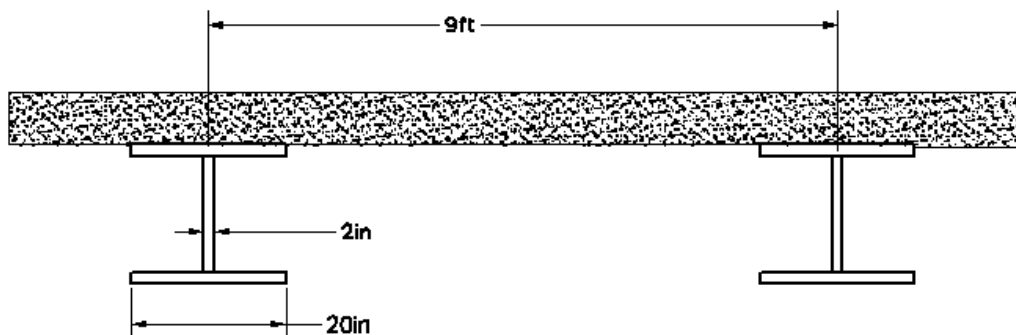
Section properties:

$b_f = 5.81$ in
 $t_f = 0.51$ in
 $d = 10.5$ in
 $t_w = 0.3$ in
 $k = 0.81$ in

- (A) 48 kips
(B) 50 kips
(C) 70 kips
(D) 75 kips

P2
160

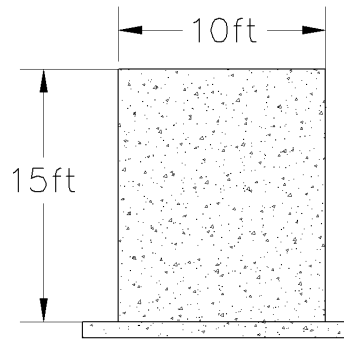
- 337.** For the bridge section shown below, what is the effective span length ' L_{eff} ' of the concrete deck to be used for empirical design per AASHTO LRFD specification?



- (A) 7 ft
(B) 7.667 ft
(C) 8.083 ft
(D) 9 ft

P2
161

- 386.** What is the axial load capacity of the cantilevered concrete wall as shown below? The thickness of the wall is 12in . Assume the resultant of loads is located within the middle third of the wall. $f'_c = 5000$ psi. Wall reinforcement complies to minimum requirements by ACI 318-14.

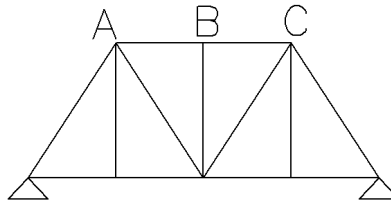


- (A) 250 kips
- (B) 310 kips
- (C) 480 kips
- (D) 520 kips

P2
162

- 409.** The truss shown below is subjected to external loads. After analysis, the internal forces in member BC for two service loading conditions are as follows:

- Case A) Dead load = 5 kip tension
- Live load = 15 kip tension
- Case B) Dead load = 5 kip tension
- Live load = 18 kip compression



What is the most economical double angle section that satisfies the design requirement?

Span of the truss = 20ft and unbraced length of member BC = 8ft.

Gr. 36 Steel

- (A) 2 L 2X2X1/8
- (B) 2 L 2X2X1/4
- (C) 2 L 2X2X5/16
- (D) 2 L 2X2X3/8

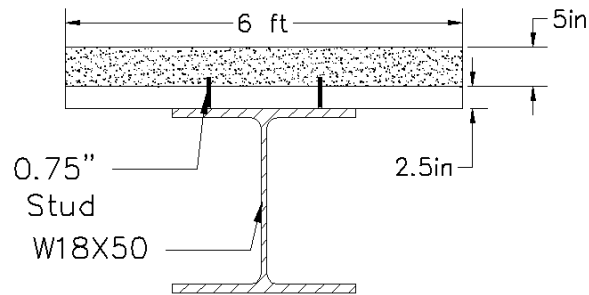
P2
163

- 419.** An angle brace is connected to a gusset plate using a combination of high strength bolts and fillet welds. The design permits sharing of loads between bolts and welds. The maximum tension load acting on the brace is 100 kips. What is the minimum load that the bolts alone must be able to withstand?

- (A) 33 kips
- (B) 50 kips
- (C) 66 kips
- (D) 75 kips

P2
164

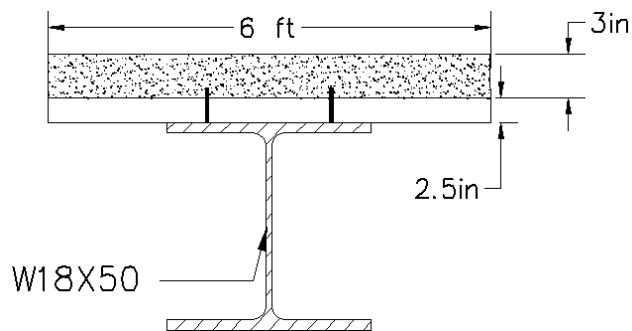
- 399.** A simply supported composite beam with span 28 ft has ribs of deck perpendicular to the beam span as shown below. The effective width of the slab is 6 ft and two 0.75 inch diameter headed studs are used per rib. For full composite action, what is the total number of studs required? The beam is subjected to uniformly distributed load. Nominal shear strength of stud, $Q_n = 15$ kips. W18X50 Gr. 50 Slab $f'_c = 4$ ksi



- (A) 36
(B) 49
(C) 98
(D) 116

P2
165

- 402.** A simply supported composite steel beam subjected to uniformly distributed load has two studs per rib as shown below. The total number of studs in the span is 64 and the nominal shear strength of each stud is 16.28 kips. What is the allowable moment capacity of the composite section? Strength of concrete slab $f'_c = 4250$ psi



- (A) 319 kip*ft
(B) 446 kip*ft
(C) 513 kip*ft
(D) 544 kip*ft

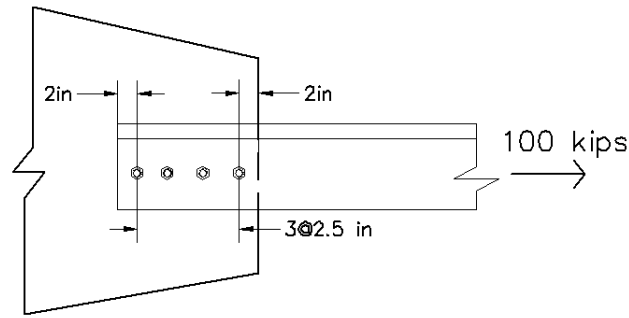
P2
166

- 440.** A pin connected hanger plate is connected to a support beam. The hanger plate is subjected to tension due to pick loads. The diameter of the pin is 3 inches and the thickness of the plate is 1.5 in. What is the allowable bearing strength of the connection? $F_y = 36$ ksi and $F_u = 58$ ksi for plates.

- (A) 120.5 kips
(B) 124.4 kips
(C) 145.8 kips
(D) 194.4 kips

P2
167

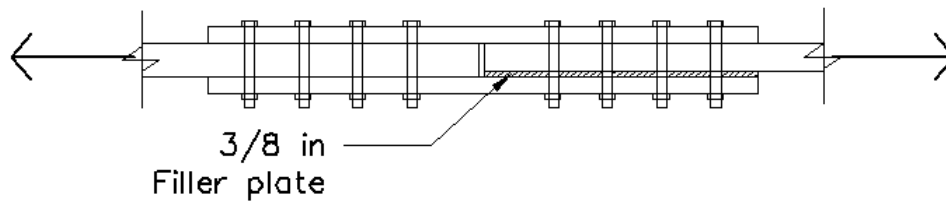
- 426.** Determine the required minimum thickness of a gusset plate that connects an angle with four bolts as shown below. Assume tensile yielding of the plate governs. Yield strength of plate is 36 ksi and tensile strength is 58 ksi. The loads shown are service loads.



- (A) 1/2 in
- (B) 5/8 in
- (C) 3/4 in
- (D) 1 in

P2
168

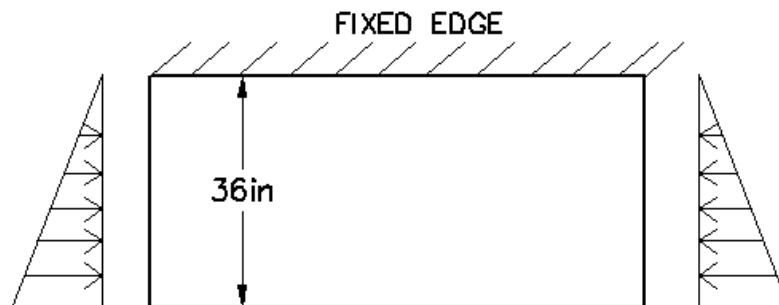
- 429.** A tension splice is constructed for two plates with different thickness as shown below. To make up for the gap, a 3/8 in thick filler plate is used. The fasteners used are 5/8 inch diameter A307 grade bolts. What is the allowable shear strength of an individual bolt to be used for the connection?



- (A) 4.14 kips
- (B) 7.87 kips
- (C) 8.29 kips
- (D) 12.5 kips

P3
169

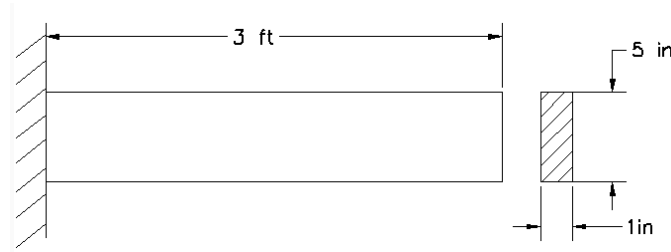
- 284.** A plate is rigidly welded at the top along the length as shown below and loaded along the thickness with a uniformly varying load. The thickness of the plate is 1 in and $E = 29000000$ psi. What is the nominal elastic buckling stress? The yield strength of the plate is 50 ksi.



- (A) 32.5 ksi
- (B) 42 ksi
- (C) 48.5 ksi
- (D) 50 ksi

P3
170

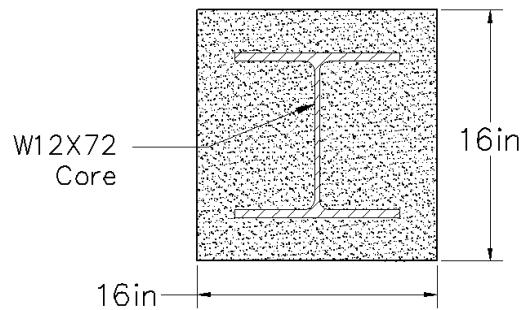
- 327.** A cantilevered horizontal bar of cross section 5in X 1in is cantilevered with $L = 3\text{ ft}$ as shown below. The yield strength of the material is 50 ksi and $E = 29000\text{ ksi}$. What is the allowable flexural strength of the bar? Assume $C_b = 1$.



- (A) 12.1 kip*ft
(B) 13.2 kip*ft
(C) 14.9 kip*ft
(D) 15.5 kip*ft

P3
171

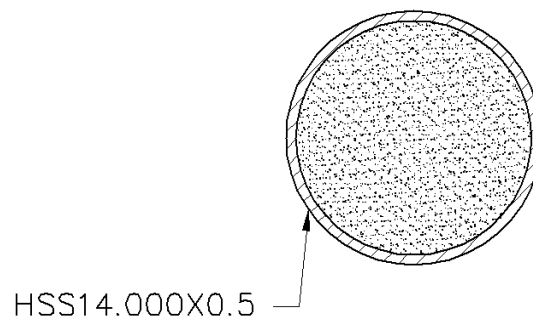
- 397.** The critical elastic buckling load of a composite 16in X 16in concrete column with A992 Gr. 50 W12X72 steel core is 850 kips. What is the allowable compressive load on the composite section? $f'_c = 4000\text{ psi}$



- (A) 372 kips
(B) 425 kips
(C) 600 kips
(D) 758 kips

P3
172

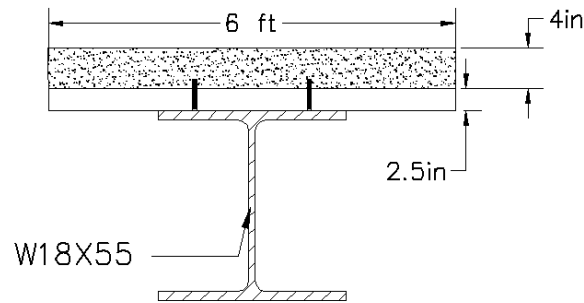
- 398.** Gr 46 HSS 14.000X0.5 is filled with concrete ($f'_c = 4\text{ ksi}$). What is the allowable compressive strength of the composite column section? Elastic critical buckling load of the section is 600 kips. Modulus of elasticity of steel, $E_s = 29000\text{ ksi}$



- (A) 263 kips
(B) 318 kips
(C) 450 kips
(D) 526 kips

P3
173

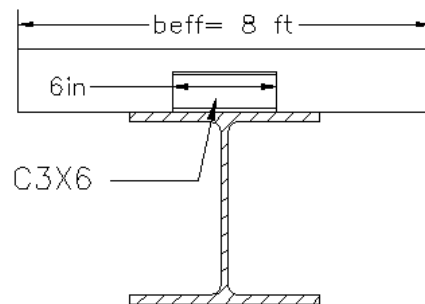
- 403.** A simply supported composite steel beam subjected to uniformly distributed load has two studs per rib as shown below. The total number of studs in the span is 40 and the nominal shear strength of each stud is 16.8 kips. Full composite action is not developed due to the limited number of studs. What is the moment of inertia of the effective composite section developed? Strength of concrete slab $f'_c = 5550$ psi



- (A) 1319 in⁴
(B) 1631 in⁴
(C) 1970 in⁴
(D) 2980 in⁴

P3
174

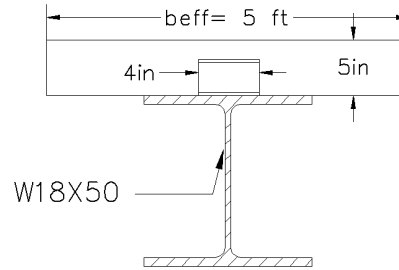
- 406.** A simply supported composite steel beam subjected to uniformly distributed load has Channel C3X6 welded to the top of the steel beam and embedded in a concrete slab with effective width 8 ft. The compressive strength of the concrete is 4 ksi and the modulus of elasticity is 3600 ksi. If the nominal shear to be transferred by the embedded channels is 380 kips, how many channels are required over the full 15 ft span of the beam?



- (A) 2
(B) 4
(C) 6
(D) 8

P3
175

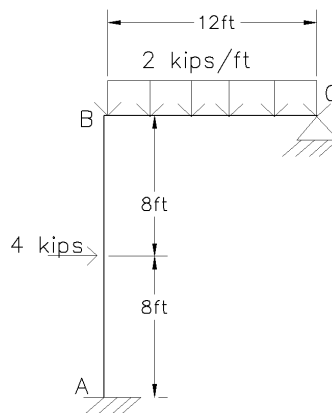
- 407.** A composite steel beam has four inch long channels, welded throughout the span to transfer horizontal shear. The beam is subjected to a factored dead load moment of 240 kip*ft and a factored live load moment of 400 kip*ft. The distance between the top of the steel beam and the concrete compression flange force is 4 in. Which of the following is most nearly the total nominal shear carried by the channels due to the applied moments? The capacity of the steel beam and concrete slab is sufficient.



- (A) 325 kips
- (B) 430 kips
- (C) 515 kips
- (D) 760 kips

P4
176

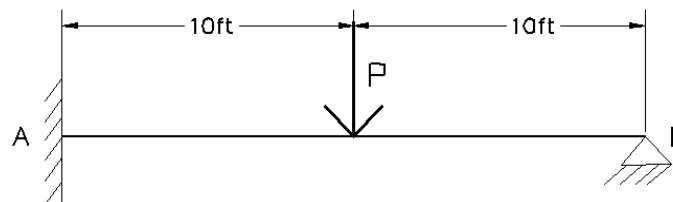
- 154.** For the framing configuration below, what is the moment reaction under column AB? The modulus of elasticity is constant. Moment of inertia of beam BC = 800 in⁴ and column AB = 550 in⁴.



- (A) 1.2 kip*ft
- (B) 2.3 kip*ft
- (C) 3.5 kip*ft
- (D) 5.8 kip*ft

P4
177

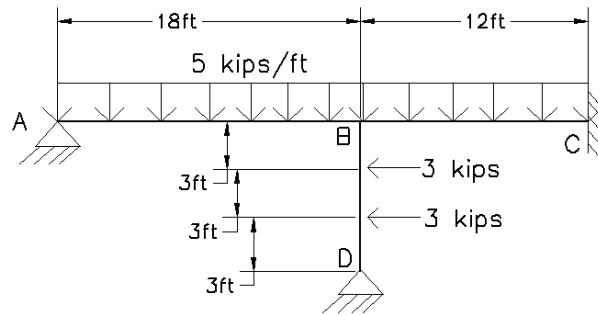
- 204.** A propped cantilever beam is shown below. The plastic moment capacity of the beam section is 100 kip*ft. What is the value of the collapse load P when the beam is analyzed via plastic analysis?



- (A) 28 kip
- (B) 30 kip
- (C) 32 kip
- (D) 34 kip

P4
178

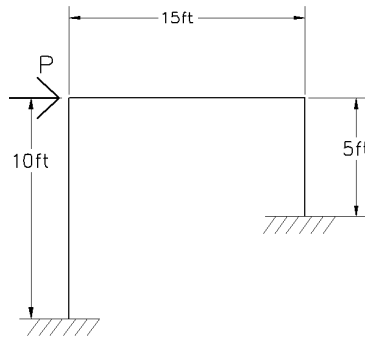
- 155.** For the framing configuration below, what is the reaction moment at point C? The material for the whole framing system is homogeneous. Moment of inertia of beam ABC = 700 in^4 and Moment of inertia for column DB = 1100 in^4 .



- (A) 23.2 kip*ft
- (B) 31.43 kip*ft
- (C) 38.24 kip*ft
- (D) 55.8 kip*ft

P4
179

- 205.** For the frame shown below, the plastic moment capacity of the section is 150 kip*ft. What is the value of lateral collapse load P?



- (A) 60 kip
- (B) 70 kip
- (C) 80 kip
- (D) 90 kip